



Why Great C++ Libraries Fail (And How to Fix That)

MATEUSZ PUSZ



20
26



Question from a workshop participant

"Which project should I contribute to learn Modern C++?"

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It depends...

The Current State of C++ Libraries

We have "Bleeding Edge" templates, **constexpr**, and concepts, but we have "Leading Edge" frustration.

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Technical excellence $\neq$ Project excellence

The "Fortress" Library

THE SYMPTOMS



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 - "This is the correct design. Period."
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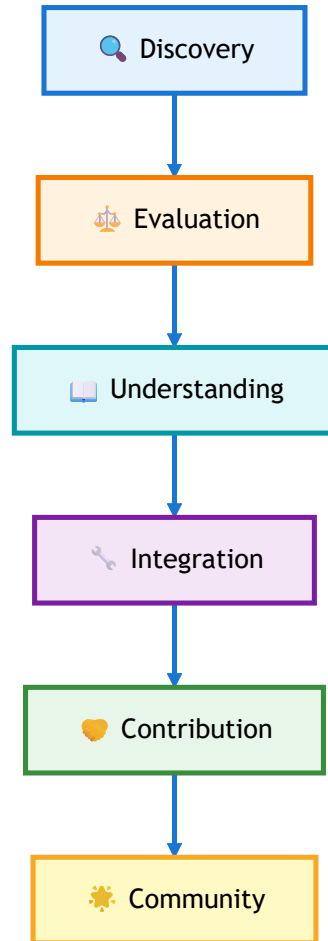
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 - "If you don't understand why, you shouldn't be contributing."
- **No clear path for contributors**
 - "Just look at the code to see how it works."
 - No **CONTRIBUTING.md**, no first issues, ...



The User Journey



- 1 Can users find your project?
- 2 Do they trust it works?
- 3 Can they learn it?
- 4 Can they actually use it?
- 5 Do they want to help?
- 6 Does it sustain itself?

STAGE 1: DISCOVERY

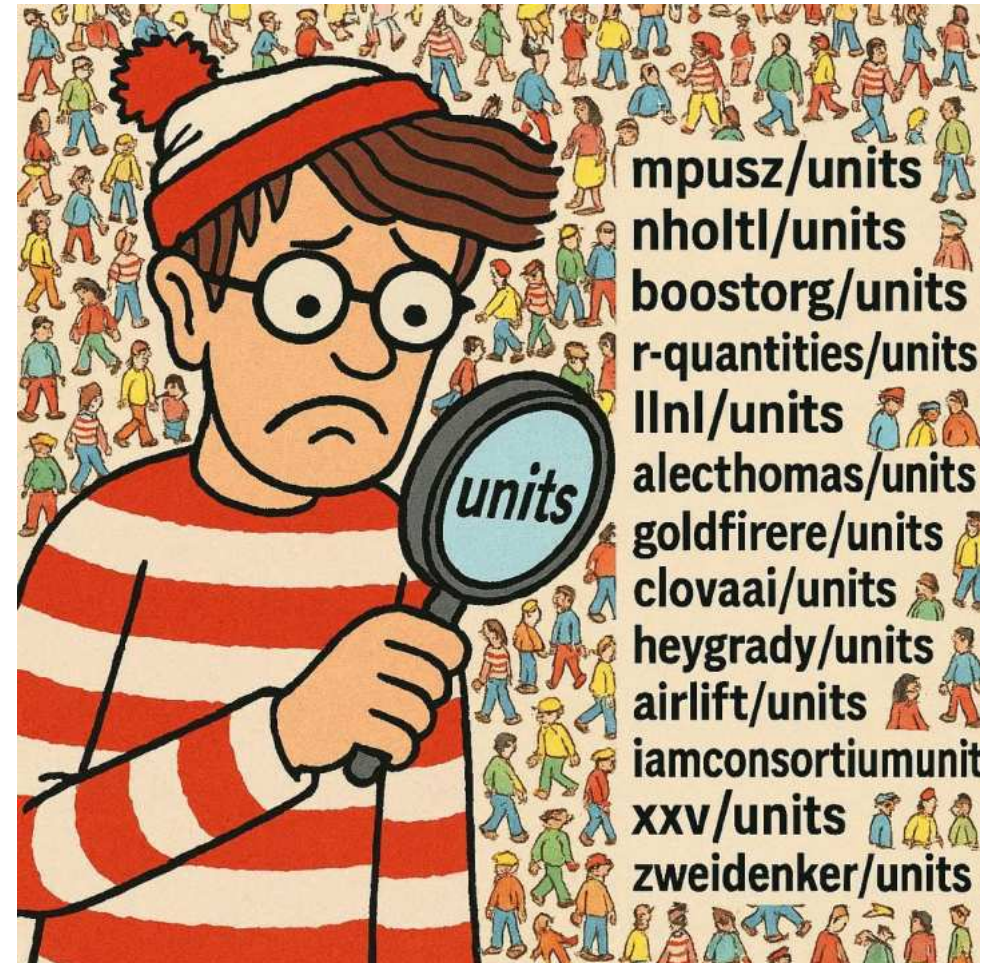
CAN USERS FIND YOUR PROJECT?

TIP 1

DISCOVERABILITY: DON'T BE A HIDDEN GEM

The Generic Name Problem

- GitHub is full of `units`, `http`, `json` libraries
- My old library: `mpusz/units` competed with `nholthaus/units`

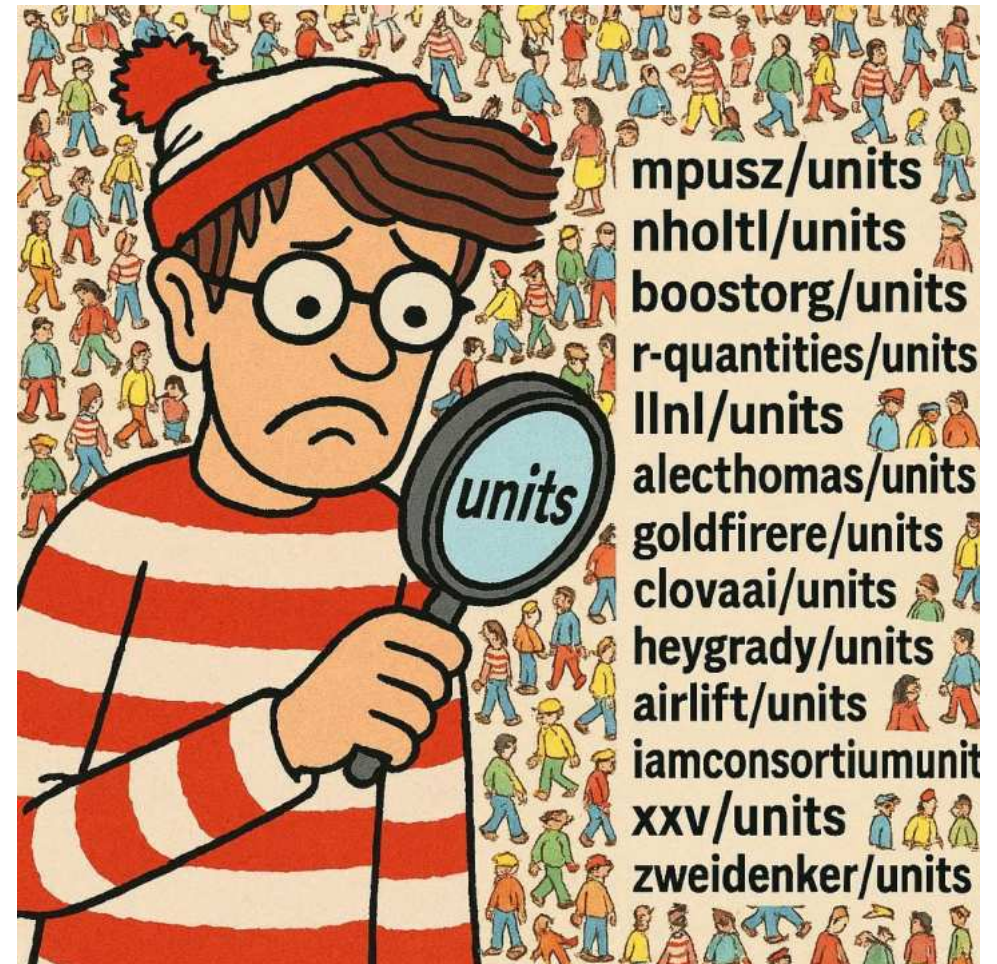


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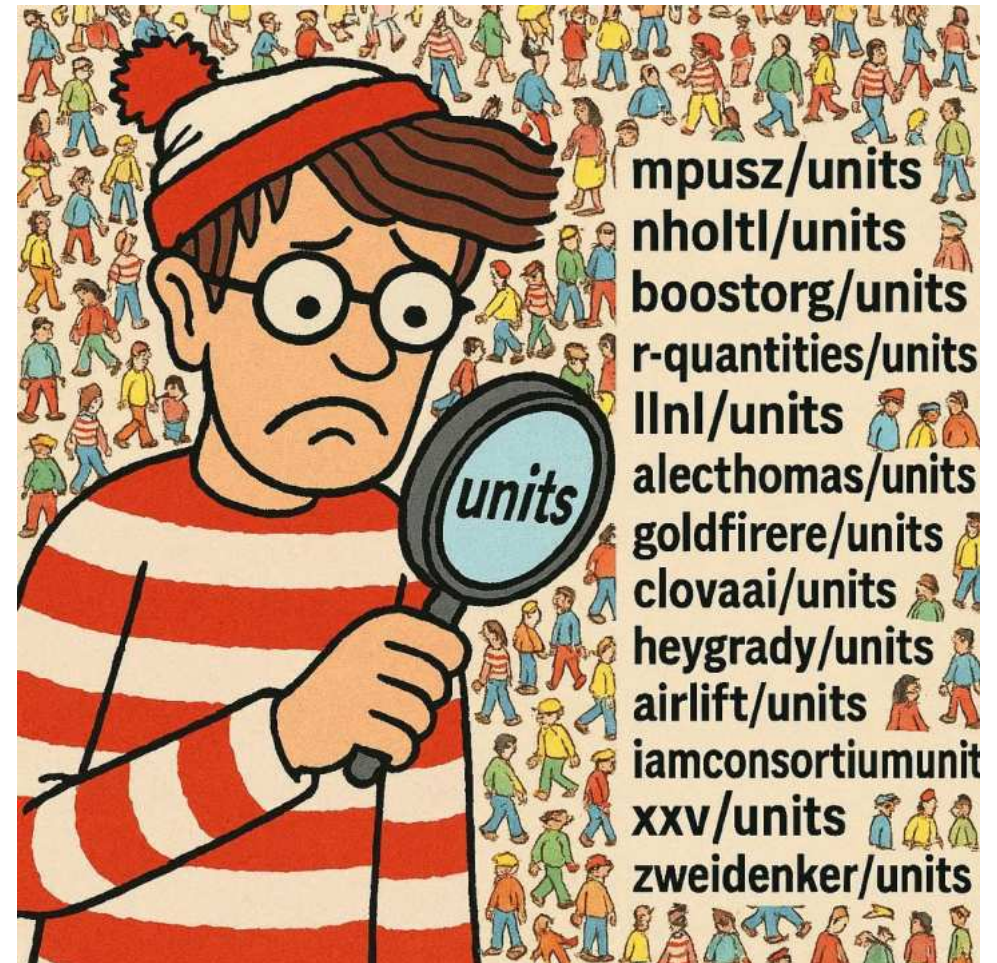
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LESSON

- Generic names create discovery and branding problems



Naming is a Technical Decision

Avoid generic names like `units`, `linalg`, `http`.

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THE MP-UNITS BLUEPRINT

- **GitHub Repo:** `mpusz/mp-units`
- **CMake Target:** `mp-units::mp-units`
- **Conan Package:** `mp-units`
- **Namespace:** `mp_units`
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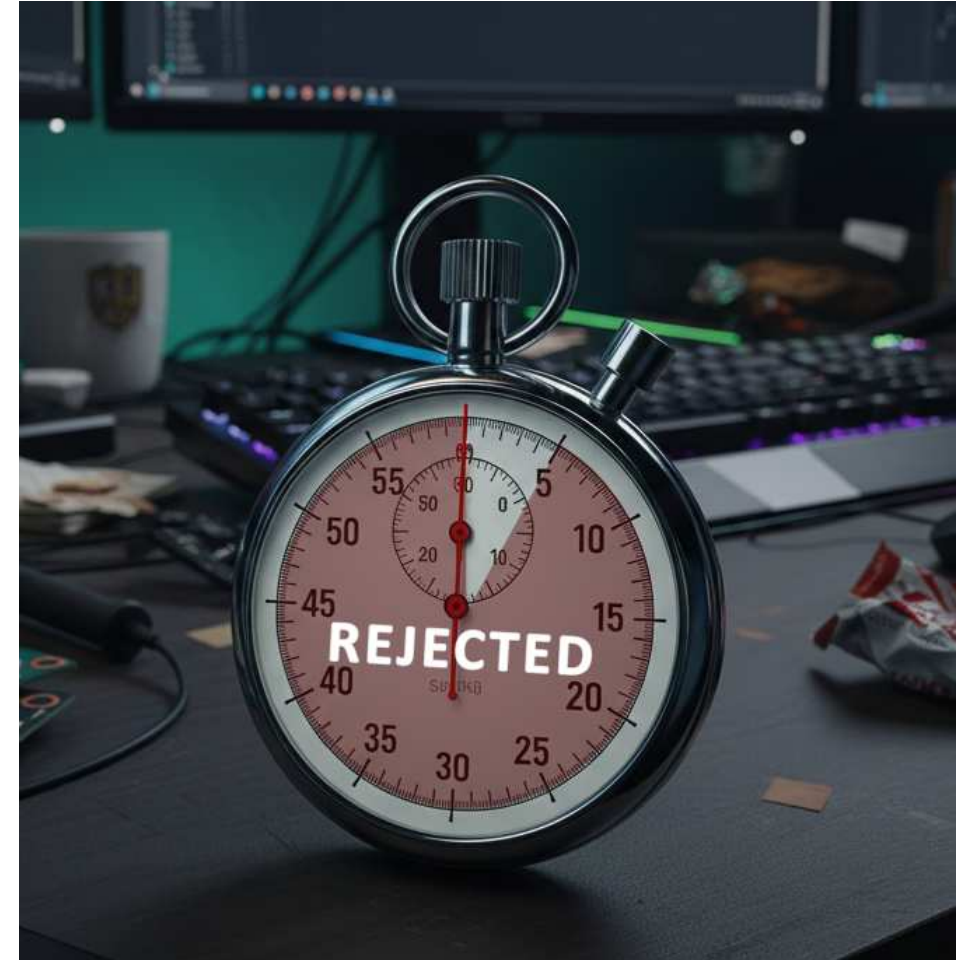
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No ambiguity, easy to search, consistent branding.

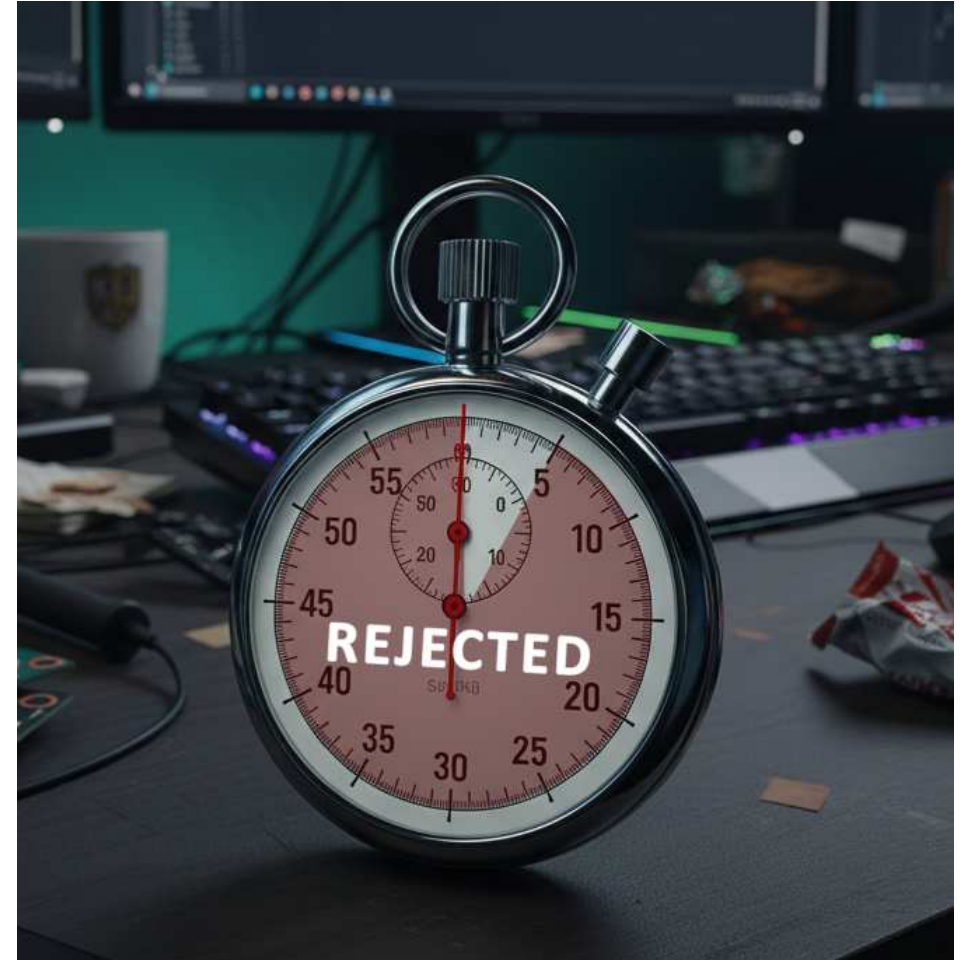
The 5-Second README Test



The 5-Second README Test

1 Logo & Badges

- Build status, coverage, license, version



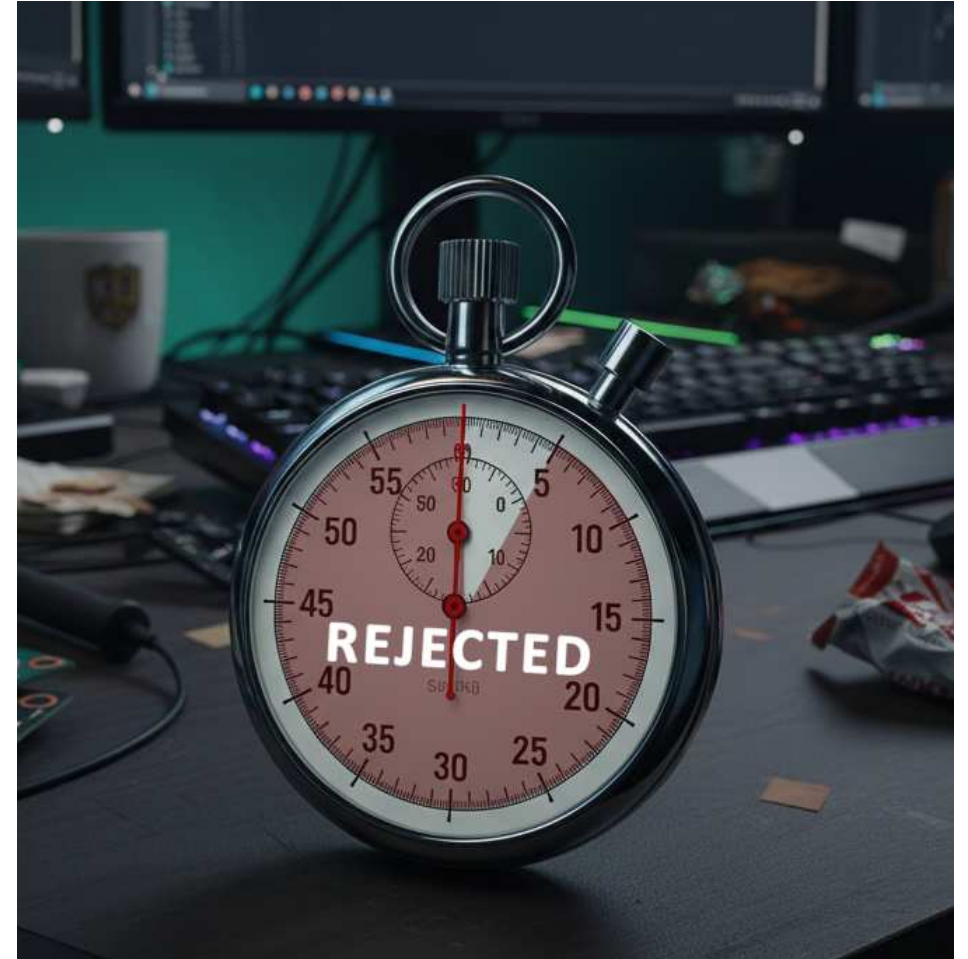
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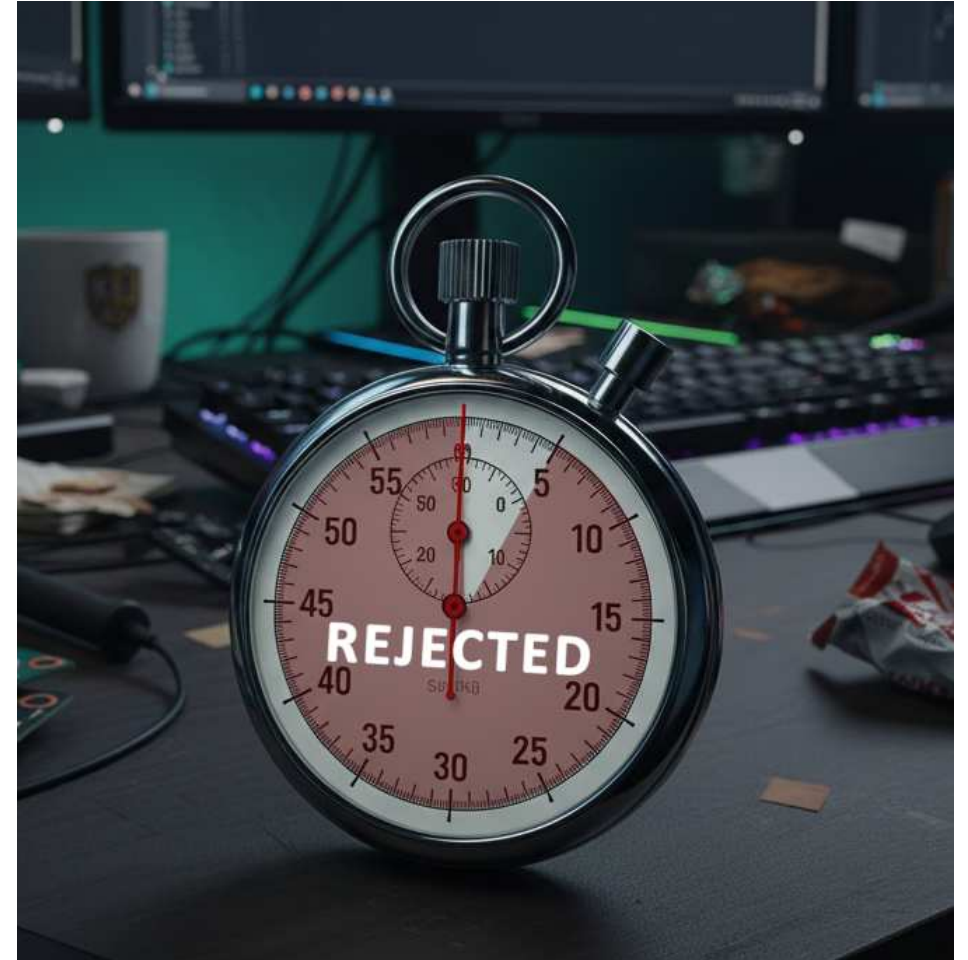
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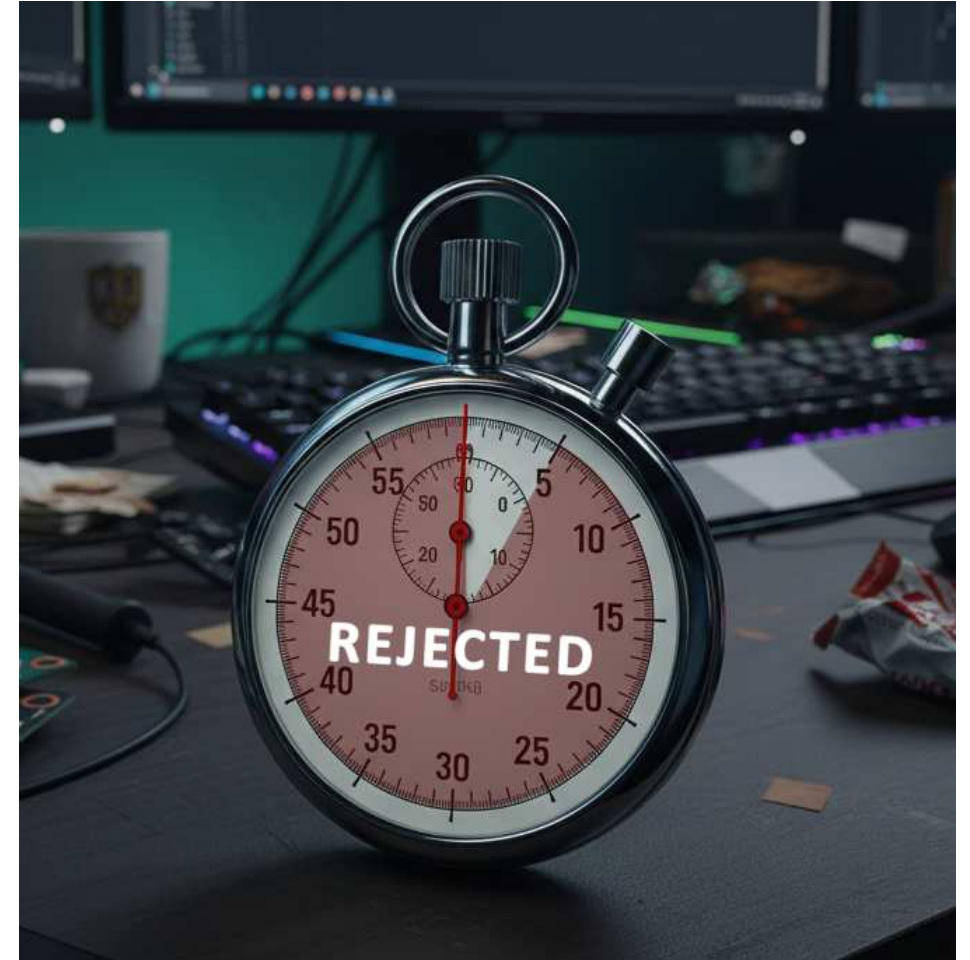
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4 Live Demo

- Compiler Explorer / interactive link



Bad vs. Good Pitch

 BAD

"A units library for C++."

Bad vs. Good Pitch

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✓ GOOD

"**mp-units** is a Modern C++ (C++20 and later) library providing the full spectrum of compile-time safety for domain-specific quantities and units, from dimensional analysis to quantity kind safety, built on the ISO 80000 International System of Quantities (ISQ). It is a candidate for C++29 standardization ([P3045](#)), your chance to shape the future of C++."

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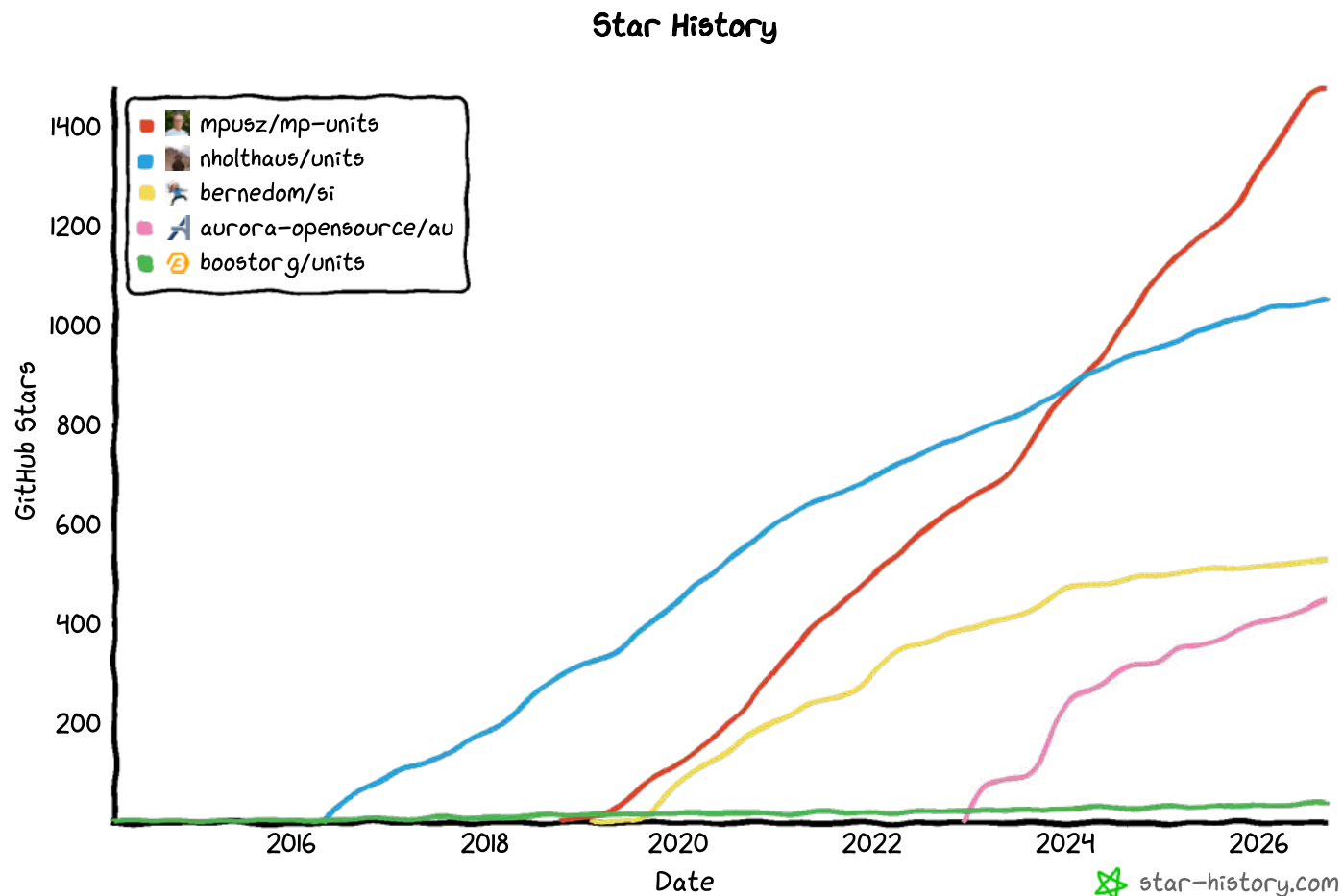
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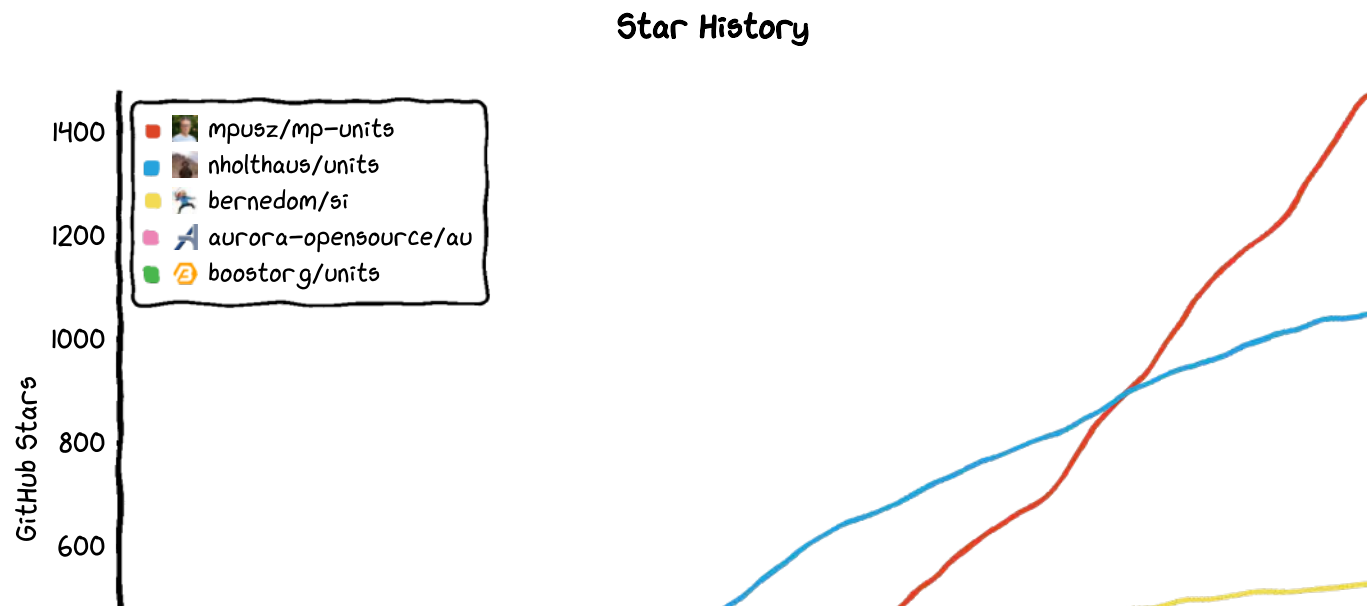
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Key Difference: Specific, verifiable, differentiating claims.

Tracking Your Reach



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Stars aren't just "vanity metrics" - they are a proxy for community trust and discoverability. Inflection points reveal what worked - use it to sharpen your strategy.

The Discoverability Blueprint

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CHECKLIST

- ☒ Unique, [searchable](#) project name
- ☒ [Consistent naming](#) (namespace = CMake = package)
- ☒ README with badges showing ["lights are on"](#)
- ☒ [Code example](#) in first screen
- ☒ Clear [value proposition](#)
- ☒ [LICENSE file](#) at root + SPDX identifier in every source header
- ☒ [SECURITY.md](#) - private vulnerability disclosure path
- ☒ [Track](#) your reach: [Star History](#)

STAGE 2: EVALUATION

DO THEY TRUST IT WORKS?

TIP 2

THE "GREEN MASTER" PHILOSOPHY

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Portability is not a "future goal". It's a daily requirement.

The CI/CD Pipeline

GitHub Actions is the gold standard for Open Source.

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COMPILERS

- GCC 12-15
- Clang 16–21
- Apple Clang 16
- MSVC 14.4

ARCHITECTURES

- x86-64, arm64

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CONFIGURATIONS

- **C++ std:** 20/23
- **Formatting:** fmtlib vs **std::format**
- **Contracts:** gsl-lite, ms-gsl, none
- **Build:** Debug/Release
- **Plus:** modules, import std, freestanding

Be the Most Restrictive User of Your Library

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- -Wall -Wextra -Wpedantic
- -Wconversion, -Wsign-conversion
- -Wshadow, -Wold-style-cast
- -Wcast-qual, -Wnull-dereference
- MSVC: /W4 /permissive-
- **Warnings as errors** in CI

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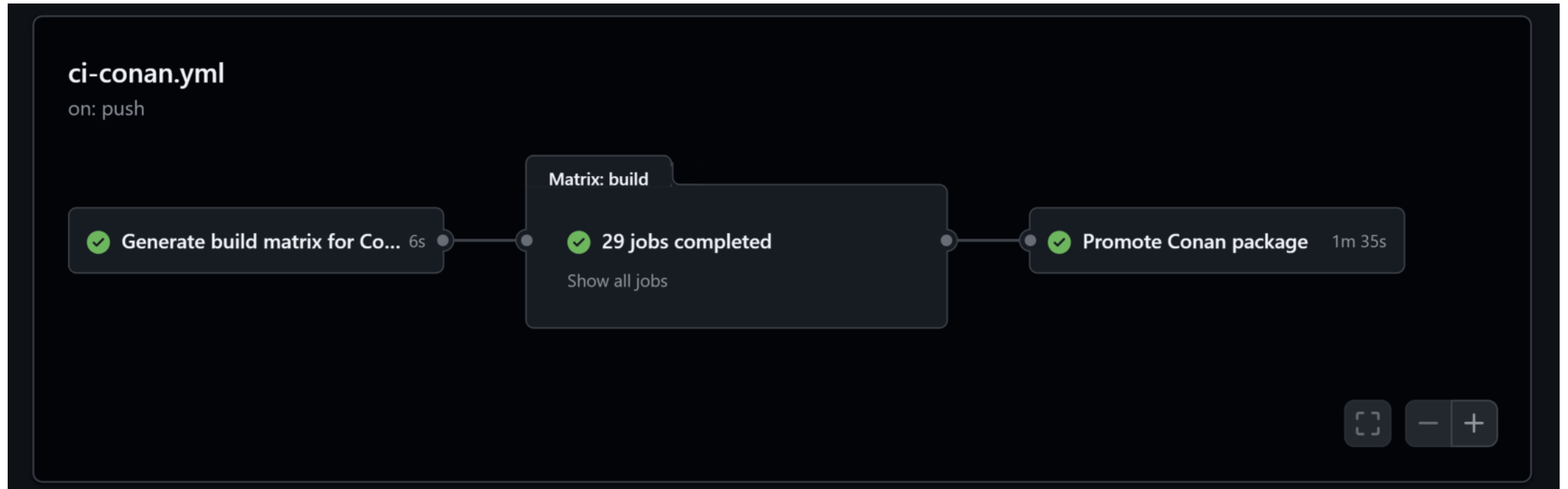
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WHY IT MATTERS

- Users may run stricter flags than you do
- Warnings from *your* headers on *their* build is a terrible first impression
- You never get to explain it

Unit testing pipeline



Unit testing pipeline



The Problem: Testing all combinations = hundreds of jobs, timeouts, quota exhaustion. **The Solution:** Fuzz Your Build Matrix.

Fuzz Your Build Matrix

Generate build matrix for Conan CI summary

 Build Matrix

Random seed: 1125297464
Total jobs: 29

Toolchain	C++	Formatting	Contracts	Build	Modules	import_std	no_crtp	freestanding
Apple Clang 16.1	20	std-format	gsl-lite	Debug	—	—	—	—
Apple Clang 16.1	23	std-format	gsl-lite	Debug	—	—	—	—
Clang-16 (x86-64)	20	fmtlib	gsl-lite	Debug	—	—	—	—
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EACH VECTOR CATCHES DIFFERENT BUGS

- **Modules:** Missing **export**
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- **Freestanding:** Accidental **<iostream>**
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Random seed: 100%
reproduction possible.

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▼ Coverage statistics

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toolchain=Clang-16 (x86-64): 2
toolchain=Clang-17 (x86-64): 2
toolchain=Clang-18 (arm64): 3
toolchain=Clang-18 (x86-64): 4
toolchain=Clang-20 (x86-64): 3
toolchain=Clang-21 (x86-64): 3
toolchain=GCC-12: 2
toolchain=GCC-13: 2
toolchain=GCC-14: 2
toolchain=GCC-15: 2
toolchain=MSVC 14.4: 2
std=20: 20
std=23: 9
build_type=Debug: 22
build_type=Release: 7
contracts=gsl-lite: 21
contracts=ms-gsl: 3
contracts=none: 5
cxx_modules=False: 19
cxx_modules=True: 10
std_format=False: 8
std_format=True: 21
import_std=False: 26
import_std=True: 3
no_crtp=False: 22
no_crtp=True: 7
freestanding=False: 29
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Clang-18 (x86-64)	20	std-format	gsl-lite	Debug	—	—	—	—
Clang-18 (x86-64)	20	std-format	gsl-lite	Debug	■	—	—	—
Clang-18 (x86-64)	23	std-format	none	Release	■	■	■	—
Clang-20 (x86-64)	20	std-format	gsl-lite	Debug	—	—	—	—
Clang-20 (x86-64)	20	std-format	gsl-lite	Debug	■	—	—	—
Clang-20 (x86-64)	23	std-format	none	Debug	■	■	■	—
Clang-21 (x86-64)	20	std-format	gsl-lite	Debug	—	—	—	—
Clang-21 (x86-64)	20	std-format	gsl-lite	Debug	■	—	—	—
Clang-21 (x86-64)	23	std-format	none	Debug	■	■	■	—
GCC-12	20	fmtlib	gsl-lite	Debug	—	—	—	—
GCC-12	20	fmtlib	ms-gsl	Release	—	—	—	—
GCC-13	20	std-format	gsl-lite	Debug	—	—	—	—
GCC-13	23	std-format	gsl-lite	Release	—	—	—	—

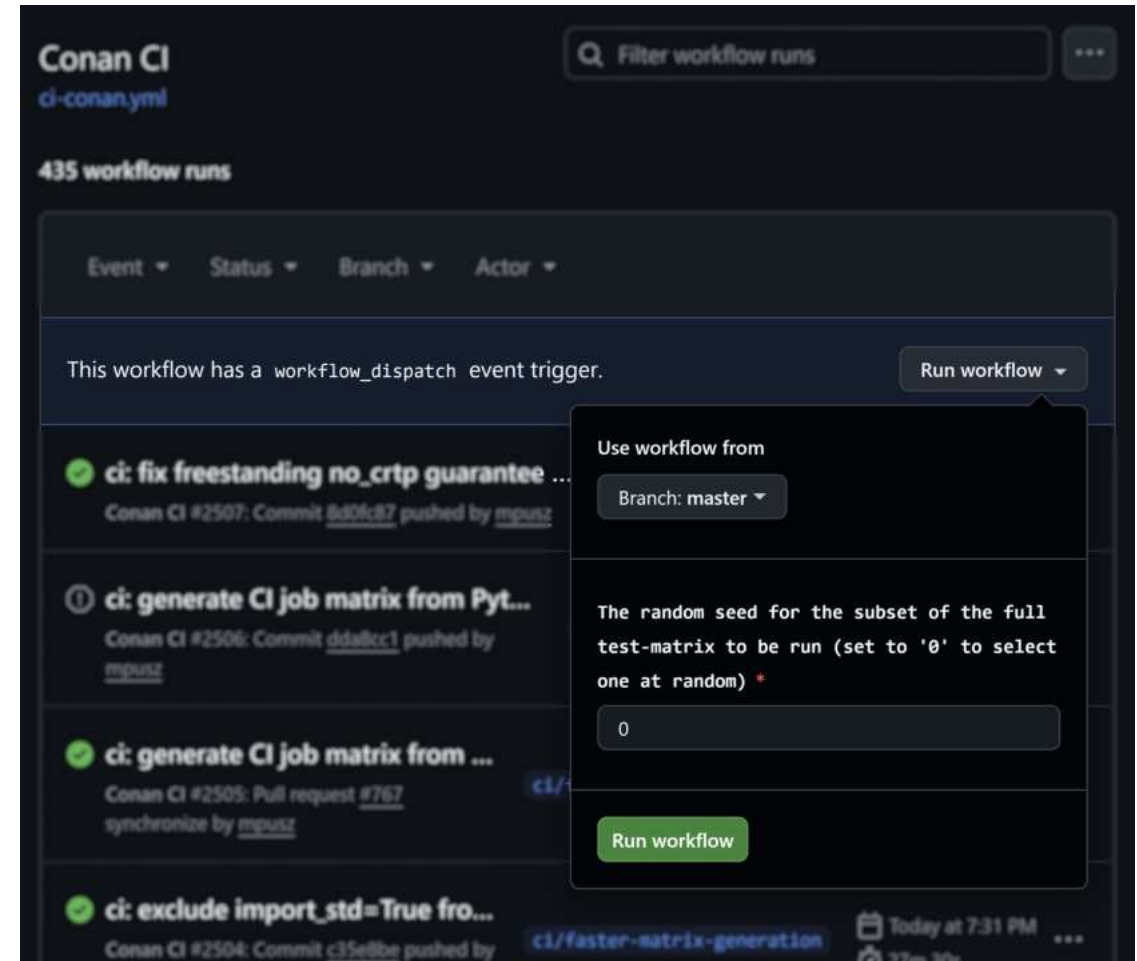
▼ Coverage statistics

```
toolchain=Apple Clang 16.1: 2
toolchain=Clang-16 (x86-64): 2
toolchain=Clang-17 (x86-64): 2
toolchain=Clang-18 (arm64): 3
toolchain=Clang-18 (x86-64): 4
toolchain=Clang-20 (x86-64): 3
toolchain=Clang-21 (x86-64): 3
toolchain=GCC-12: 2
toolchain=GCC-13: 2
toolchain=GCC-14: 2
toolchain=GCC-15: 2
toolchain=MSVC 14.4: 2
std=20: 20
std=23: 9
build_type=Debug: 22
build_type=Release: 7
contracts=gsl-lite: 21
contracts=ms-gsl: 3
contracts=none: 5
cxx_modules=False: 19
cxx_modules=True: 10
std_format=False: 8
std_format=True: 21
import_std=False: 26
```

Minimum coverage guarantee: Each config option appears at least twice.

Random, but Reproducible

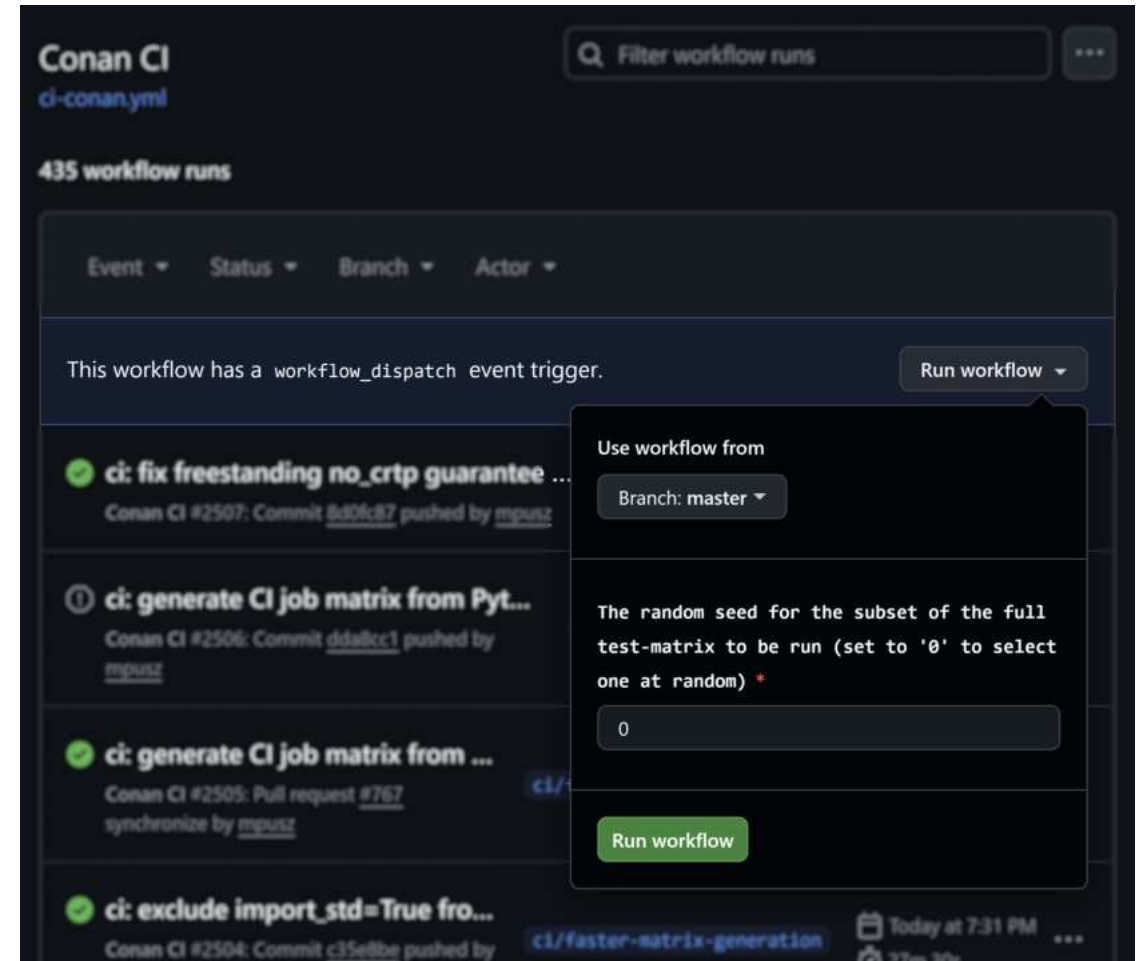
- **Random seed:** GitHub Actions
`workflow_dispatch` input
 - *Specify seed* → reproduce exact failing configuration
 - *Seed = 0* → generate new randomized matrix (printed in logs)



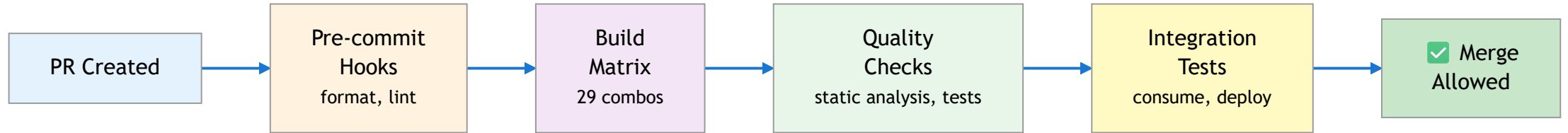
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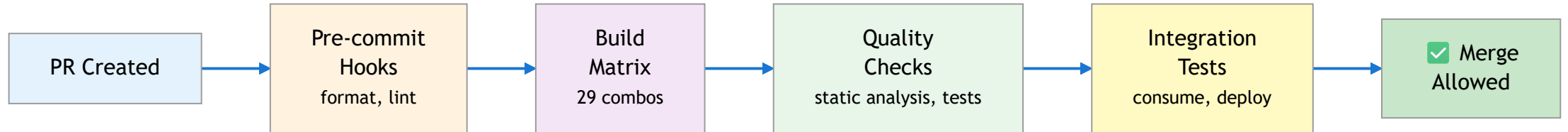
Result: Broad coverage without combinatorial explosion.



The Quality Gate - Complete Flow



The Quality Gate - Complete Flow



The Complete Defense:

- 🛡️ **6 layers** of validation before merge
- 🚫 **No single point of failure** - each layer catches different issues
- ✅ **High confidence** - if all green, it's production-ready
- 🎯 **Trust earned** - users see "all checks passed" and believe it

Performance: Prove It, Track It

A shared runner cannot compare timings across runs. It can still count.

Performance: Prove It, Track It

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ON A SHARED RUNNER

✗ Wall clock, runtime or compile time

- Identical work, no library in it at all: 32 ms and 23 ms under one hostname
- It measures the machine, not the change
- Belongs on a quiet machine, asked occasionally, not on every change

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✓ Deterministic, for a pinned toolchain

- Instantiation counts: clang's **-ftime-trace**, granularity 0 or you count nothing
- Emitted code bytes, not total binary size
- Pin the compiler, the standard and the corpus, or the number moves

Performance: Prove It, Track It

A shared runner cannot compare timings across runs. It can still count.

PRACTICAL OPTIONS

-  **Count, don't time** - gate on instantiations and emitted bytes, on every change
-  **Instrument** - CodSpeed or Valgrind: count instructions, not seconds
-  **Relative comparison** - cancels differences between machines, but not varied workload

What "All Checks Passed" Means to Users?

LOWER RISK

- **Fewer surprises** - Edge cases caught before you hit them
- **Faster debugging** - Issues reproduced quickly with same matrix
- **Better support** - Maintainer can validate your bug reports



STAGE 3: UNDERSTANDING

CAN THEY LEARN IT?

TIP 3

DOCUMENTATION AS PROSE

The Documentation Paradox

How many of you enjoy writing documentation?

The Documentation Paradox

How many of you enjoy writing documentation?

How many of you appreciate good documentation in the 3rd Party Software?

"We Have Plenty of Documentation!"



README.md Is Not Enough

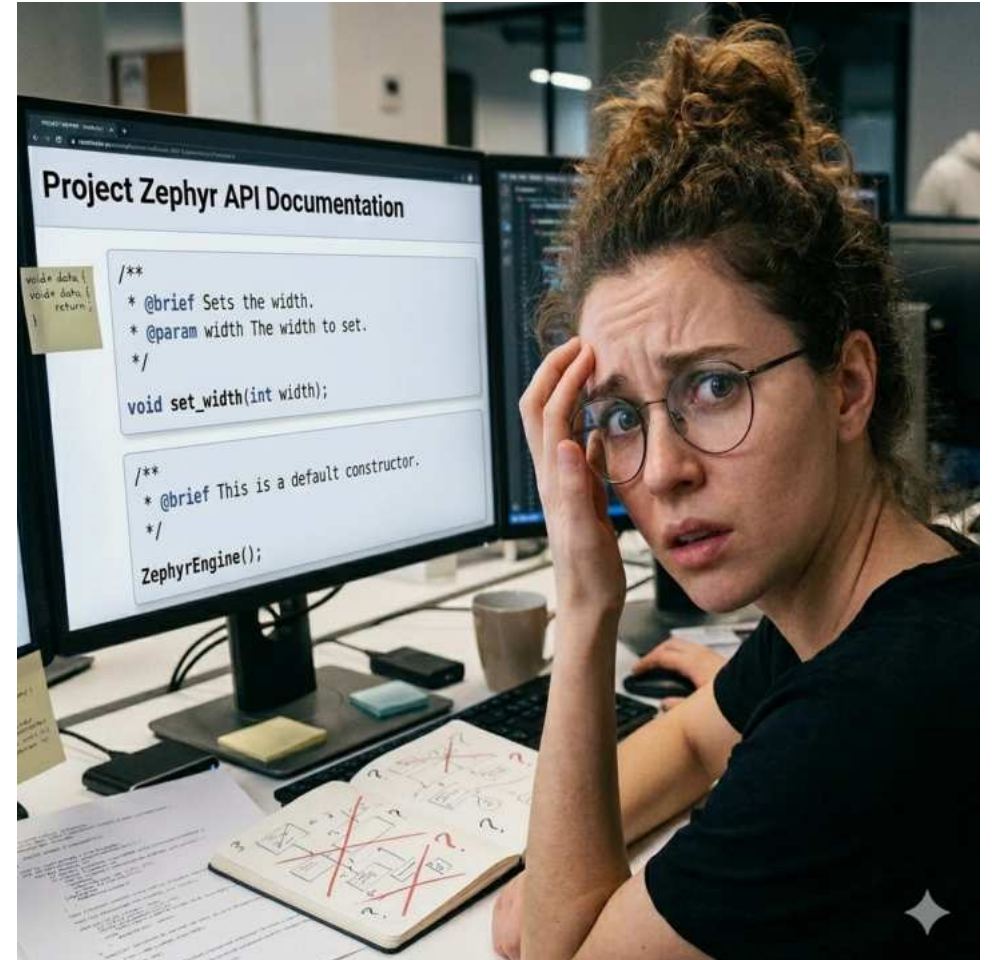
Lots of C++ projects do not provide any other documentation besides README.md.

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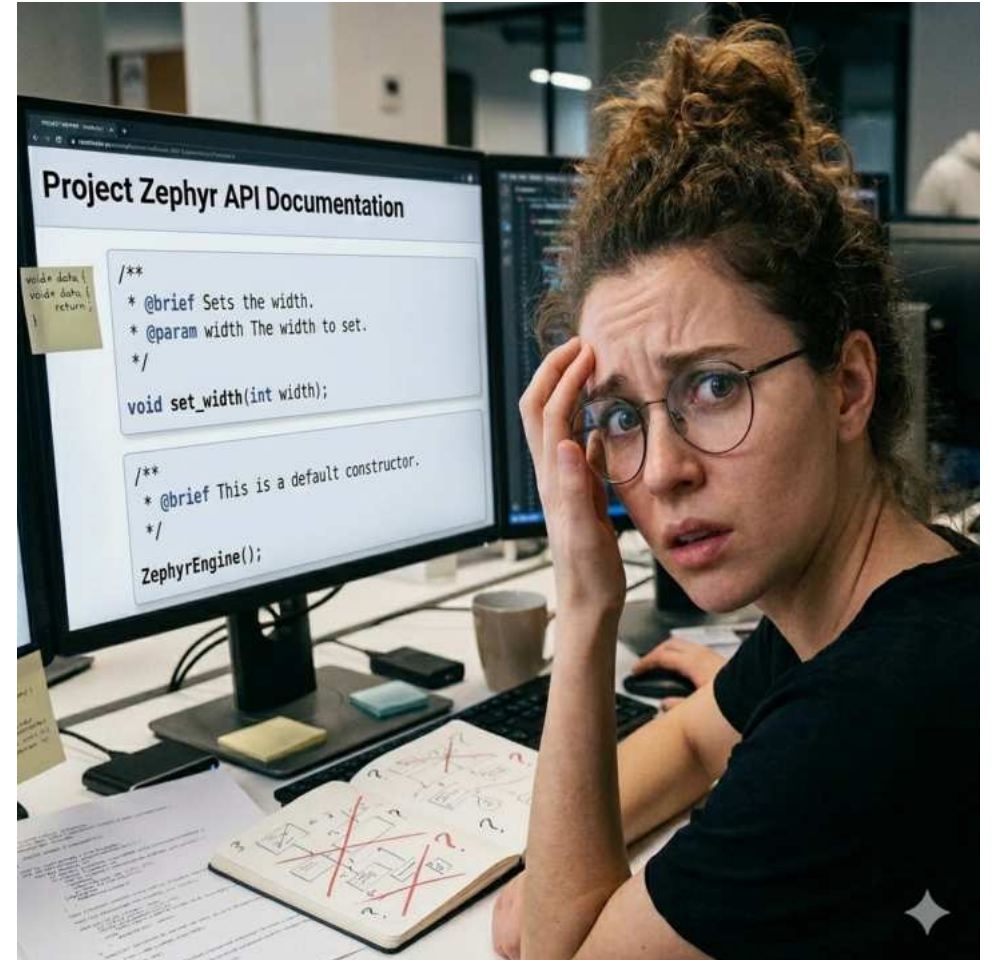
Let's just run Doxygen on the current code and publish the outcome... No!

The Doxygen Trap



The Doxygen Trap

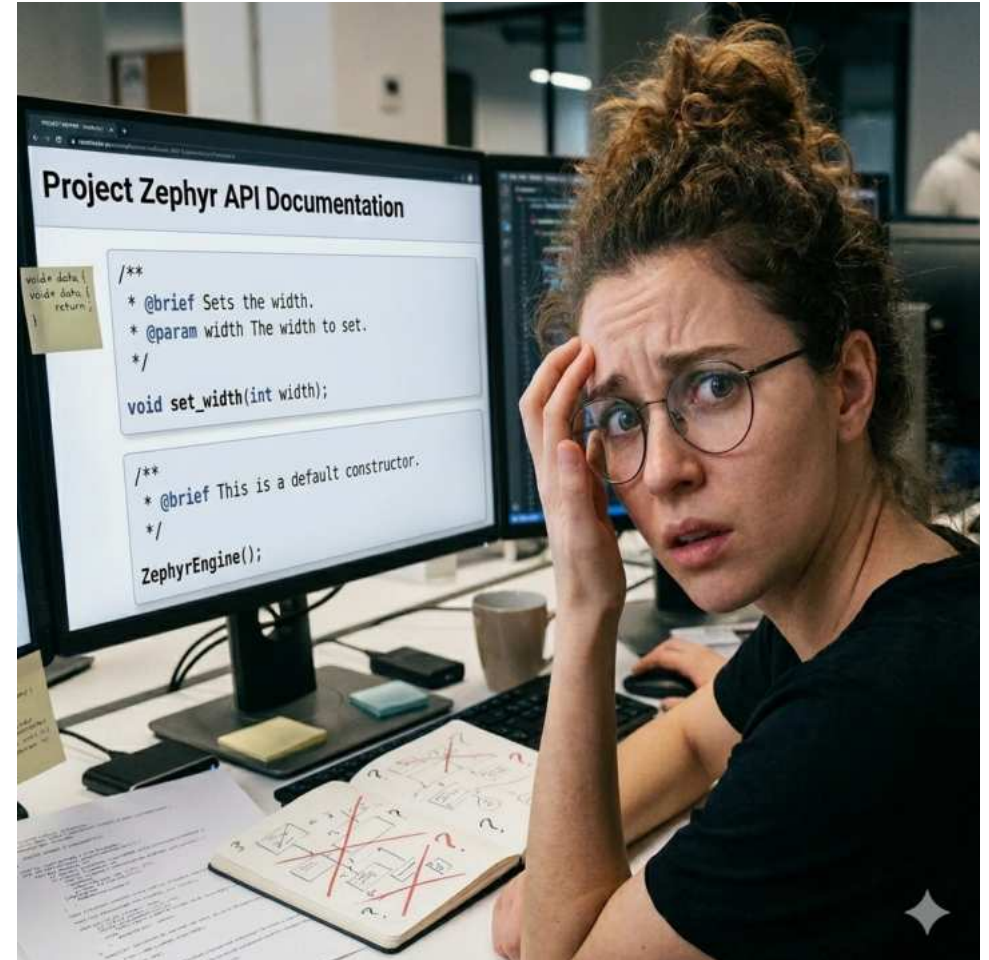
Doxygen encourages
"Inventory Documentation".



The Doxygen Trap

Doxygen encourages
"Inventory Documentation".

```
/**
 * @brief Sets the width.
 * @param width The width to set.
 */
void set_width(int width);
```



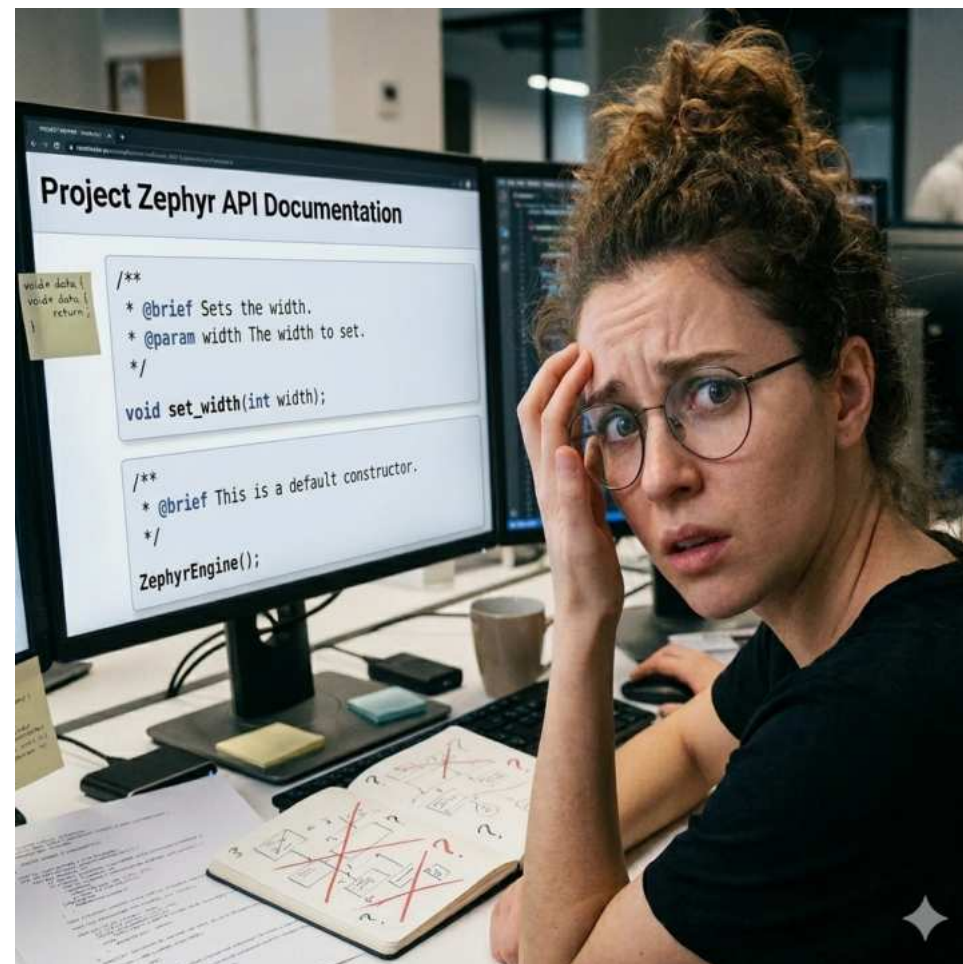
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THE TAUTOLOGY PROBLEM

- Tells you *what* is there
- Doesn't explain *why* or *how* to use it
- 1,000 pages of docs, not helpful for new users



Reference vs. Tutorial



DOXYGEN

- A dictionary
- Great when you know the word you look for
- Useless for learning the framework
- Often auto-generated from raw code with no custom descriptions 😬

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PROSE DOCUMENTATION

- A textbook
- Guides you through concepts
- Explains the architecture and rationale
- Provides a curated learning path

It is not boring at all

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Writing documentation might be a fascinating design-related exercise!

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Show off 🎉

- You spent *weeks* on this feature
- Documentation is your stage

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- "How do I explain this?" → "Why is it this hard?"
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Repeat 🔁

- Design → Implement → Document → Redesign
- Each cycle tightens the product

The Diátaxis Framework

Systematic approach to technical documentation (diataxis.fr).

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QUADRANT	USER SAYS	FOCUS
 Tutorials	"I want to learn"	Learning-oriented
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Most doxygen-based documentation address only the fourth quadrant.

mp-units Homepage

 **mp-units** HEAD 

 Search

 mpusz/mp-units
v2.5.0 1.4k 119

[Home](#) [Getting Started](#) [Tutorials](#) [User's Guide](#) [Workshops](#) [How-to Guides](#) [Examples](#) [Reference](#) [Blog](#) [Release Notes](#)

Welcome to mp-units!

mp-units is the only Modern C++ (C++20) library providing the full spectrum of compile-time safety for physical quantities and units – from dimensional analysis and unit conversions to **quantity kind safety** – built on the ISO 80000 International System of Quantities (ISQ).

The library source code is hosted on [GitHub](#) with a permissive [MIT license](#).

C++ modules

Header files

```
#include <iostream>
#include <print>
import mp_units;

using namespace mp_units;

inline constexpr struct smoot final : named_unit<"smoot", mag<67> * usc::inch> {} smoot;

int main()
{
    constexpr quantity dist = 364.4 * smoot;
    std::println("Harvard Bridge length = {:.1f} ({:N[.1f]}, {:N[.2f]}) ± 1 ear",
                dist, dist.in(usc::foot), dist.in(si::metre));
}
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Output:

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Harvard Bridge length = 364.4 smoot (2834.6 ft, 620.14 m) ± 1 ear
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Try it live on Compiler Explorer

mp-units Navigation: Diátaxis in Action

The screenshot shows the mp-units website with a dark blue header. The header contains the mp-units logo, a 'HEAD' dropdown, a search bar, and repository statistics for 'mpusz/mp-units' (v2.5.0, 1.4k stars, 119 forks). Below the header is a navigation bar with links: Home, Getting Started, Tutorials, User's Guide, Workshops, How-to Guides, Examples, Reference, Blog, and Release Notes. The main content area starts with 'Welcome to mp-units!' followed by a description of the library as a Modern C++ (C++20) library for physical quantities and units, emphasizing 'quantity kind safety' and its foundation on the ISO 80000 International System of Quantities (ISQ). It also mentions that the source code is on GitHub with a MIT license. A code block shows a C++ module example for calculating the Harvard Bridge length in smoots, feet, and meters. Below the code is the output: 'Harvard Bridge length = 364.4 smoot (2834.6 ft, 620.14 m) ± 1 ear'. A button 'Try it live on Compiler Explorer' is at the bottom. A light blue box on the right, titled 'Diátaxis Categories', lists: Tutorials (Learning) with a blue circle, How-To (Goal) with a green circle, Explanation with a yellow circle, Reference with a red circle, Workshops (Hybrid) with a purple circle, Beyond Diátaxis with a grey circle, and Home/Welcome with a white circle.

mp-units HEAD

mpusz/mp-units v2.5.0 1.4k 119

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STAGE 4: INTEGRATION

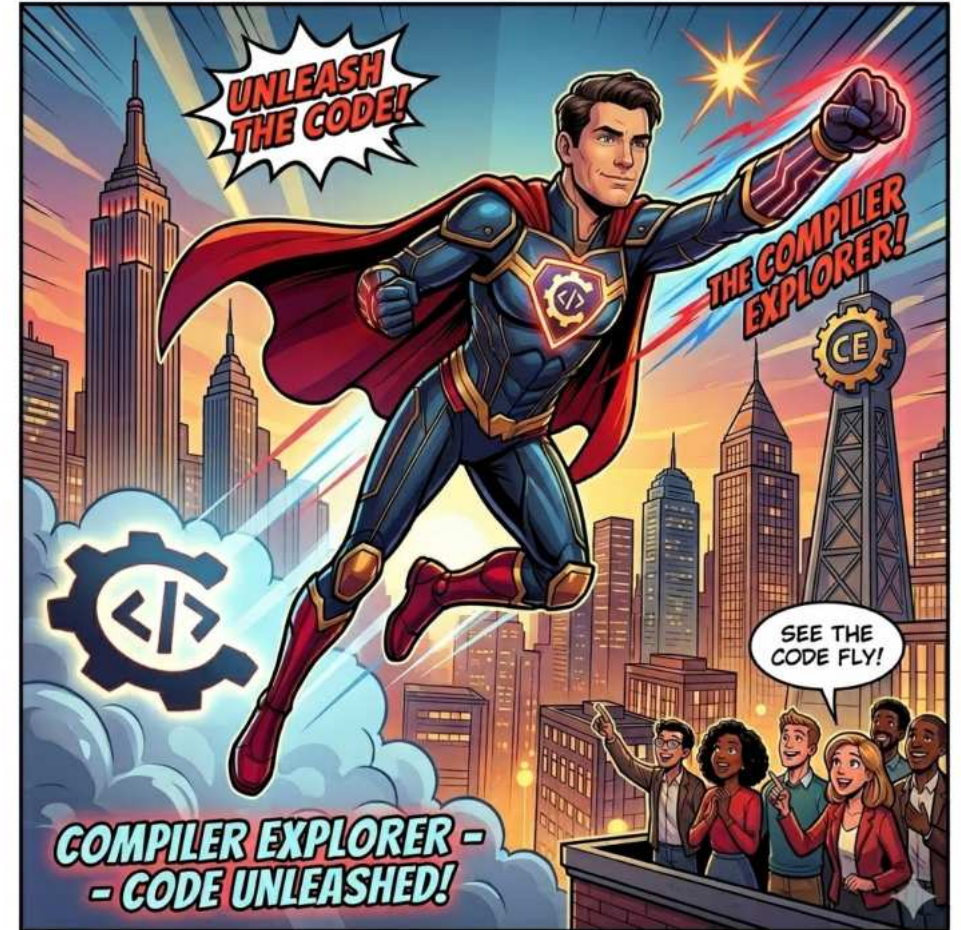
CAN THEY ACTUALLY USE IT?

TIP 4

ZERO-INSTALL PROTOTYPING

Compiler Explorer

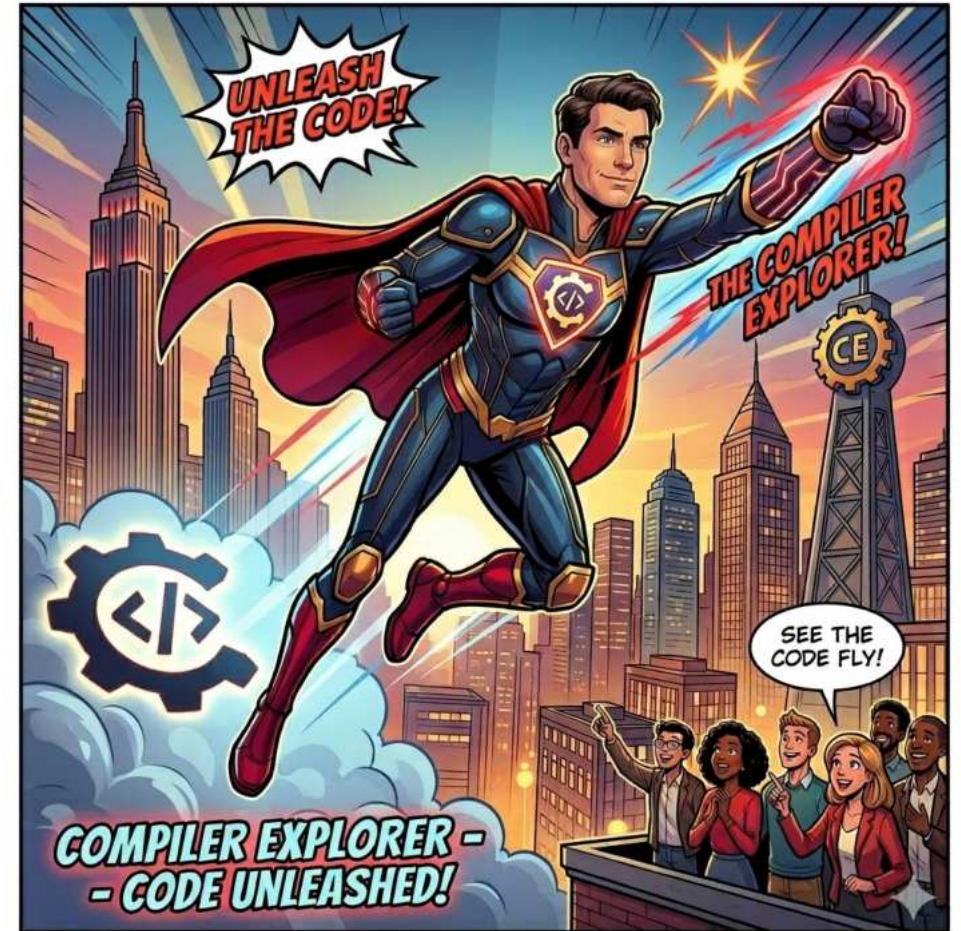
Arguably the most important C++ tool of the last decade.



Compiler Explorer

Arguably the most important C++ tool of the last decade.

- **Zero-install** - play with code before downloading anything
- **Instant sharing** - one link, no repo, no setup
- **Library support** - especially easy for header-only libraries
- **Responsive team** - reach out, they will help you add your library



Compiler Explorer: Full Lifecycle Coverage

Prototyping

- Explore ideas in seconds
- No build system needed

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- Eliminates the "it doesn't work on my machine" loop

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- Show the assembly don't just claim it
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Contributor Onboarding

- Explore the codebase without cloning or building anything
- "Here's what I want to fix" before writing a single line

Zero-Overhead Proof

- Show the assembly don't just claim it
- Let skeptics verify your abstractions cost nothing

Documentation

- "Try it out" links in README and docs
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Don't claim zero overhead - show the assembly.

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NO UNITS LIBRARY

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double ttg_s(double d_m, double v_mps)
{
    return d_m / v_mps;
}
```

MP-UNITS

```
quantity<s> ttg(quantity<m> d, quantity<m/s> v)
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```

Try it for yourself: <https://godbolt.org/z/dvaq3af1T>.

Interactive tutorials

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mp-units HEAD

Search

mpusz/mp-units
v2.5.0 1.4k 118

Home Getting Started Tutorials User's Guide Workshops How-to Guides Examples Reference Blog Release Notes

Tutorials

- Quick Start
- Working with Units
 - Unit Conversions
 - The .in() Method
 - Conversion in Expressions
 - Common Conversions
 - Function Interfaces with Automatic Conversion
 - Challenges
 - What You Learned?
- Safe and Unsafe Conversions
- Extracting Numeric Values
- Type Safety
- Affine Space

Unit Conversions

Learn to convert quantities between different units safely and explicitly.

Goal: Convert the same quantity to different units

Time: ~15 minutes

The .in() Method

To express a quantity in different units, use `.in()`:

```
1 #include <mp-units/systems/si.h>
2 #include <mp-units/systems/yard_pound.h>
3 #include <iostream>
4
5 int main()
6 {
7     using namespace mp_units;
8     using namespace mp_units::si::unit_symbols;
9     using namespace mp_units::yard_pound::unit_symbols;
10
11     // Distance in meters
12     quantity distance = 5.0 * km;
13
14     std::cout << "Original: " << distance << "\n";
15     std::cout << "In meters: " << distance.in(m) << "\n";
16     std::cout << "In miles: " << distance.in(mi) << "\n";
17     std::cout << "In feet: " << distance.in(ft) << "\n";
18
19     // Speed conversion
```

Program returned: 0

Program stdout

```
Original: 5 km
In meters: 5000 m
In miles: 3.10686 mi
In feet: 16404.2 ft

Speed: 100 km/h
```

x86_64 clang 21.1.0 118ms

Making It Work for Your Library

GETTING ADDED TO COMPILER EXPLORER

- 1 Open an issue/PR on [compiler-explorer/compiler-explorer](#)
- 2 Header-only → trivially easy
- 3 The team is **friendly and responsive**

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BEST PRACTICES

- Use **shortlinks** (godbolt.org/z/abc123) - keep docs clean
- Add "▶ **Try it on Compiler Explorer**" links to every README example
- Require a **Godbolt reproduction link** in your bug report template
- Use **interactive embeds** for doc tutorials

Making It Work for Your Library

GETTING ADDED TO COMPILER EXPLORER

- 1 Open an issue/PR on [compiler-explorer/compiler-explorer](#)
- 2 Header-only → trivially easy
- 3 The team is **friendly and responsive**

BEST PRACTICES

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Please do provide Release tags for your library → stability guarantee.

TIP 5

SEAMLESS INTEGRATION

Don't be asocial!



David Sankel: Big Projects, and CMake, and Git, Oh My!

Don't be asocial!

A library is asocial when it lacks proper means to consume it.

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✗ ASOCIAL LIBRARY

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- No CMake exported targets
- No package manager recipe
- Obscure or home-grown dependencies
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✓ SOCIAL LIBRARY

- Well-known build tools and dependencies
- **find_package()** + namespaced targets
- On [Conan Center](#) and/or [vcpkg](#)
- Out-of-source, CMake Presets support
- User's flags are respected

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- User's flags are respected

Principle: Provide the integration you wish you had as a user.

Package Management: Conan & vcpkg

conanfile.txt

```
[requires]
mp-units/2.5.0

[options]
mp-units*:std_format=True
mp-units*:contracts=gsl-lite

[generators]
CMakeToolchain
CMakeDeps
```

```
conan install . -pr gcc15 -b missing
cmake --preset conan-default
cmake --build --preset conan-release
```

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```
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```

FOR USERS

- Zero manual dependency management
- *Feature flags* become Conan options
- *Reproducible builds* across teams

Meet Users Where They Are

Not everyone uses a package manager - and that's okay.

Meet Users Where They Are

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LEGACY & CORPORATE ENVIRONMENTS

-  **Strict dependency policies** - no external package managers allowed
-  **Air-gapped networks** - no internet access at build time
-  **Existing monorepos** - vendoring is the standard

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NICHE & EXPERIMENTAL BUILDS

-  **CPM / FetchContent** - CMake-native, no extra tools
-  **Bazel, Meson, build2** - not everyone uses CMake
-  **add_subdirectory()** - simple vendoring, zero overhead

Supported Consumption Paths

- 🏆 **Conan Center** - official releases
- 🔗 **vcpkg** - ships with Visual Studio
- 🚀 **CPM / FetchContent** - no package manager
- 📁 **add_subdirectory()** - vendoring
- 🛠️ **Install + find_package()** - system-wide

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The Principle: The more paths you support, the fewer reasons to say no.

Modern CMake: Targets, Not Variables

CONSUMER'S CMakeLists.txt

```
find_package(mp-units REQUIRED)

target_link_libraries(my_app
    PRIVATE
        mp-units::mp-units
)
```

Modern CMake: Targets, Not Variables

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WHAT TARGETS GIVE YOU?

- ✓ Include paths - automatic
- ✓ Compile flags - propagated
- ✓ Transitive dependencies - resolved
- ✓ PRIVATE/PUBLIC/INTERFACE - respects user's graph
- ✓ Install rules - **find_package()** works after **cmake --install**

Modern CMake: Targets, Not Variables

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WHAT TARGETS GIVE YOU?

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- ☒ Install rules - **find_package()** works after **cmake --install**

Avoid: `include_directories()`, playing with `CMAKE_CXX_FLAGS`, changing ABI related flags, or setting any compilation warning flags.

Wide Compatibility: The C++ Reality

Feature support varies wildly across the compiler matrix.

Wide Compatibility: The C++ Reality

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C++ FEATURE	C++ VERSION	GCC	CLANG	APPLE-CLANG	MSVC
Minimum support	20	12+	16+ && !19	15-16 && !17.0	19.5+
<code>std::format</code>	20	13+	17+	16+	19.5+
C++ modules	20	None	17+	None	None
<code>import std;</code>	23	15+	18+	None	19.5+
Explicit <code>this</code> parameter	23	14+	18+	17+	19.5+
Contracts	26	16+	None	None	None

Wide Compatibility: The C++ Reality

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<code>import std;</code>	23	15+	18+	None	19.5+
Explicit this parameter	23	14+	18+	17+	19.5+

Your users are spread across every cell of this table. Meet them all!

Wide Compatibility: Choose Your Style

CONAN CONFIGURATION OPTIONS

OPTION	VALUES
<code>cxx_modules</code>	<code>True/False</code>
<code>import_std</code>	<code>True/False</code>
<code>std_format</code>	<code>True/False</code>
<code>no_crtp</code>	<code>True/False</code>
<code>contracts</code>	<code>none / std / gsl-lite / ms-gsl</code>
<code>freestanding</code>	<code>True/False</code>

Wide Compatibility: Choose Your Style

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STRATEGY

- Targeting latest compilers? → **terse modern style**
- Stuck on older toolchains → **use alternatives**
- Need portability? → **compatibility macros**
- Start portable, refactor later

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- Start portable, refactor later

All combinations tested in the CI matrix!

STAGE 5: CONTRIBUTION

DO THEY WANT TO HELP?

TIP 6

COLLABORATIVE TRIAGE

Collaborative Triage

Communication IS a feature of your library.

Collaborative Triage

Communication IS a feature of your library.

How you treat an issue defines your community.

Separate Issues from Discussions

ISSUES

- Actionable bugs
- Tasks
- Specific Feature Requests
- Security vulnerabilities

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DISCUSSIONS

- "How do I...?" (Q&A)
- Show and Tell
- Ideas
- Design Proposals / RFCs
- Brainstorming

Separate Issues from Discussions

ISSUES

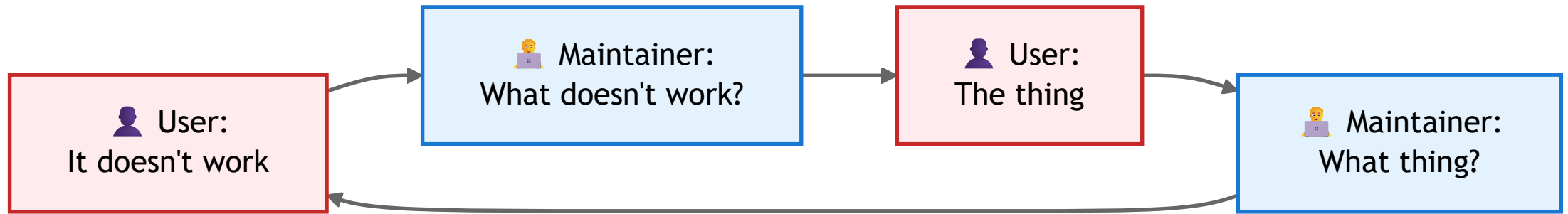
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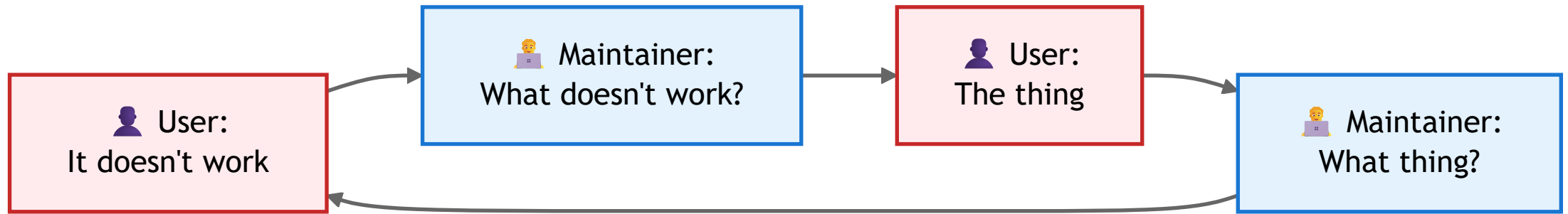
- "How do I...?" (Q&A)
- Show and Tell
- Ideas
- Design Proposals / RFCs
- Brainstorming

Enable GitHub Discussions - it's free.

The "It Doesn't Work" Loop



The "It Doesn't Work" Loop

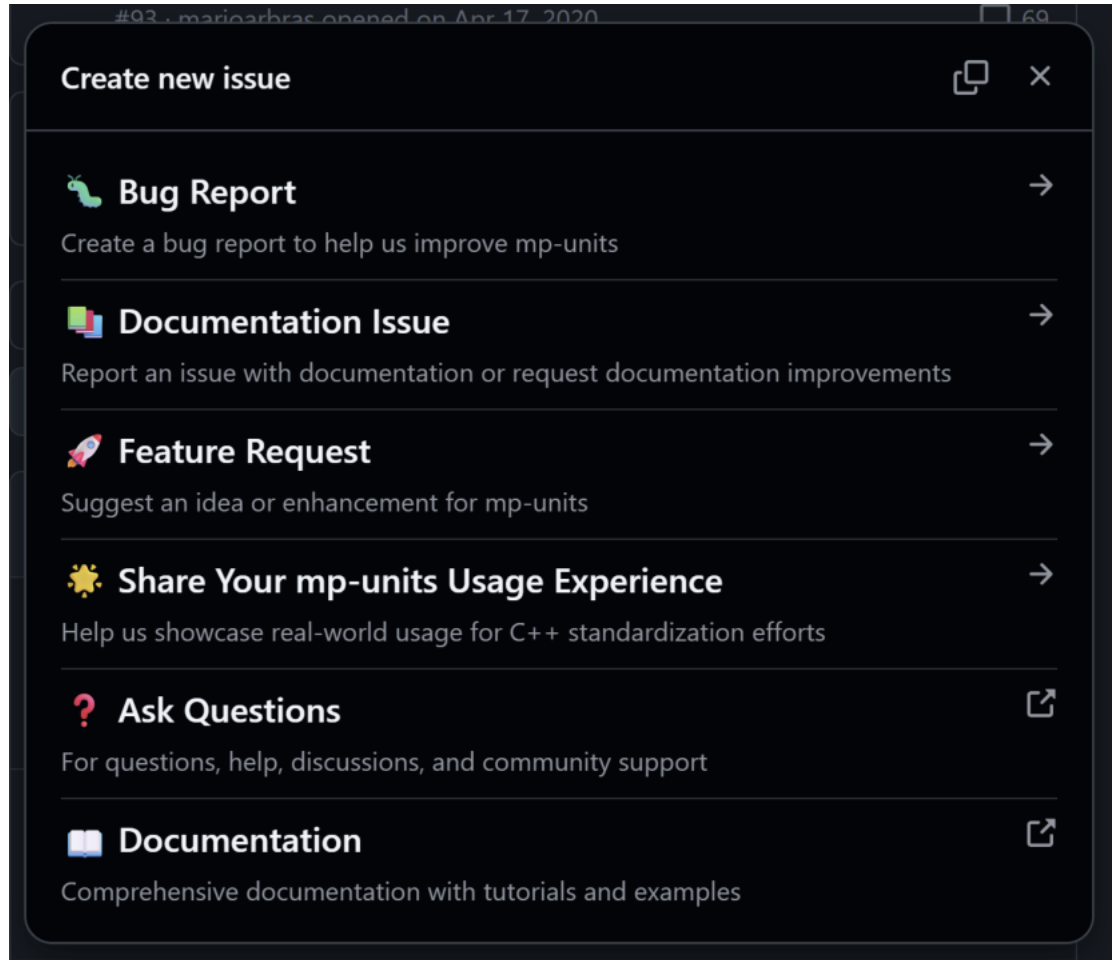


This wastes everyone's time - Stop it with templates!

Issue Templates

REQUEST

- Description
- Steps to reproduce
- **Compiler Explorer link**
- Build config
 - OS
 - Compiler
 - C++ version
 - product version
 - product features enabled



TIP 7

ONE-CLICK DEV ENVIRONMENTS

User's Scenario: "I want to fix a typo"



One-Click Dev Environments

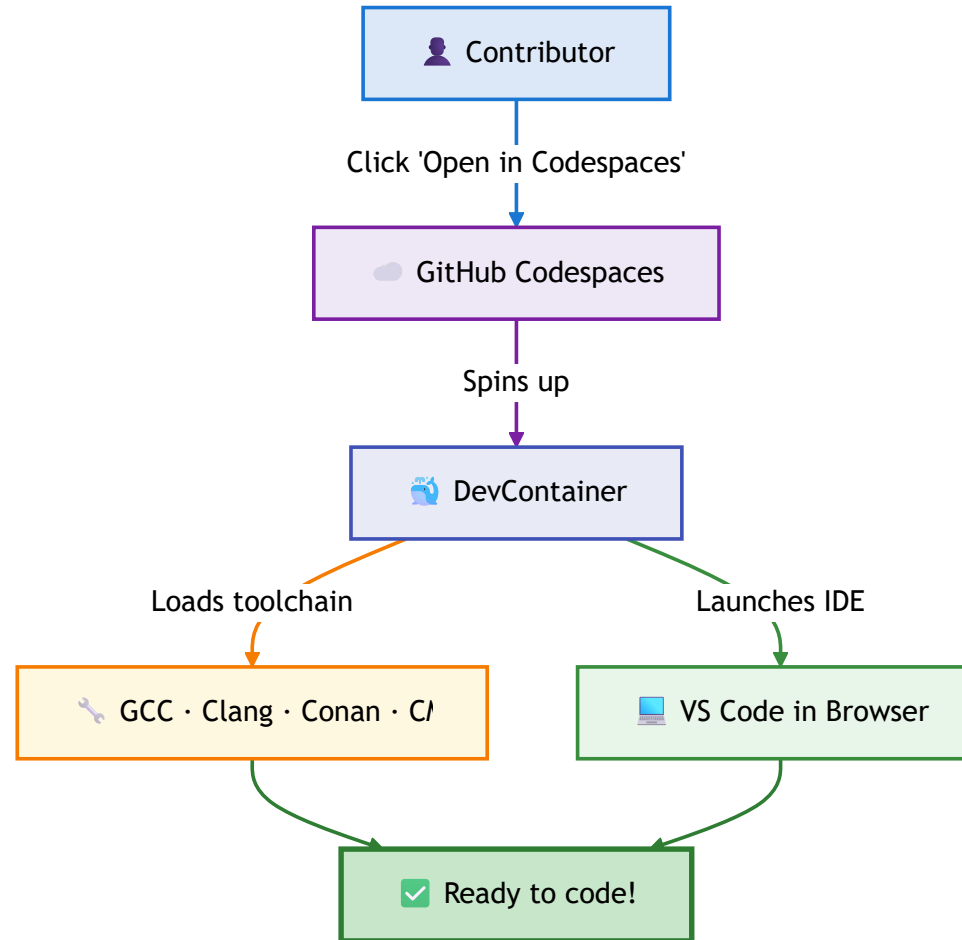
Instead of describing the development environment provide one as a container.

One-Click Dev Environments

Instead of describing the development environment provide one as a container.

Toolchain, config files, dependencies pre-installed.

GitHub Codespaces: Easy Contributing



devcontainer.json: Toolchain as Code

```
{
  "image":
    "trainiteu/mp-units:latest",
  "postCreateCommand":
    ".devcontainer/post-create.sh",
  "customizations": {
    "vscode": {
      "extensions": [
        "ms-vscode.cpptools",
        "ms-vscode.cmake-tools",
        "..."
      ]
    }
  }
}
```

WHAT'S PRE-INSTALLED

-  **Compilers:** GCC 12–15 · Clang 16–22
-  **Build:** CMake · Ninja · Conan 2.x
-  **Analysis:** clang-format · clang-tidy · IWYU
-  **Docs:** MkDocs + Material theme
-  **Profiles:** All Conan configs pre-set
-  **Cache:** Dependencies pre-compiled

devcontainer.json: Toolchain as Code

```
{
  "image":
    "trainiteu/mp-units:latest",
  "postCreateCommand":
    ".devcontainer/post-create.sh",
  "customizations": {
    "vscode": {
      "extensions": [
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        "ms-vscode.cmake-tools",
        "..."
      ]
    }
  }
}
```

WHAT'S PRE-INSTALLED

-  **Compilers:** GCC 12–15 · Clang 16–22
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Remember to test this in CI as well.

Not Just for External Contributors

💛 TEAM CONSISTENCY

- **Identical** environment for every developer
- No "works on my machine" syndrome
- New team member → productive on day 1
- Container = CI image → **local == CI**

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💻 RUN LOCALLY TOO

- Docker + VS Code Dev Containers extension
→ **"Reopen in Container"**
- Full offline development
 - No cloud usage limits
 - Better performance on local hardware

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🔬 ISOLATED PR TESTING

```
# Fresh Codespace per PR branch
conan build . -pr clang21 \
  -c user.mp-units.build:all=True -b missing
```

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# Fresh Codespace per PR branch
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```

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- Docker + VS Code Dev Containers extension
→ **"Reopen in Container"**
- Full offline development
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🔧 FULL MATRIX TESTING

```
# All compilers - similar to CI
.devcontainer/check_all.sh create
```

GCC 12–15 · Clang 16–22 · C++20/23/26

Trivial local reproduction

Clang-20 (x86-64) C++23 std::format modules=yes import_std=yes no_crtp=yes none Release summary ...

✗ Freestanding Build Failed: Clang-20 (x86-64) C++23 std::format modules=yes import_std=yes no_crtp=yes none Release

Local repro (in the provided '.devcontainer' or GitHub CodeSpaces environment):

```
conan build . -b missing -s compiler=clang -s compiler.version=20 -s compiler.cppstd=23 -s build_type=Rel
```

Job summary generated at run-time

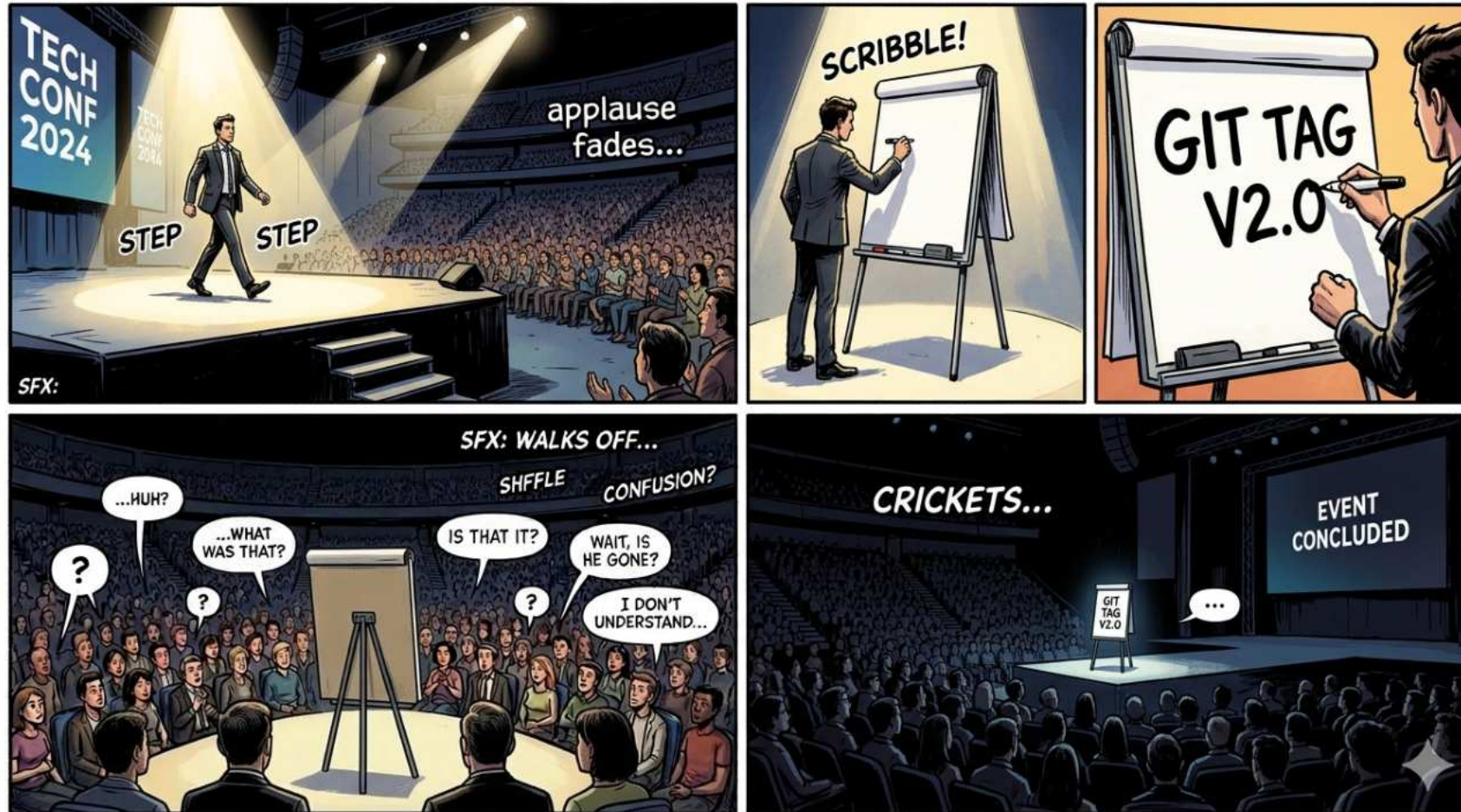
STAGE 6: COMMUNITY

DOES IT SUSTAIN ITSELF?

TIP 8

THE RELEASE BLOG

A git tag is not user communication



A git tag is not user communication

Releases are marketing events.

A git tag is not user communication

Releases are marketing events.

A tag on GitHub is not a release announcement.

Prose-Based Release Notes

- **The "Why":** Why did this version happen?
- **Highlights:** What are the 3 big things?
- **Breaking Changes:** How do I fix my code?

Communicating Breaking Changes

In C++, "breaking" is often invisible - until a user's build fails.

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In C++, "breaking" is often invisible - until a user's build fails.

IN THE CODE: `[[deprecated]]`

```
namespace [[deprecated("2.3.0: Use mp_units::iec from iec.h instead")]] iec80000 { ... }  
  
[[deprecated("2.6.0: Use si::absolute_zero instead")]]  
inline constexpr auto usc_absolute_zero = ...
```

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IN THE CODE: `[[deprecated]]`

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namespace [[deprecated("2.3.0: Use mp_units::iec from iec.h instead")]] iec80000 { ... }

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inline constexpr auto usc_absolute_zero = ...
```

IN THE RELEASE BLOG

-  **breaking change**  marker on every breaking item
- **Before / After** code tabs showing the exact migration
- **!!! note** callouts with removal timeline

The Blog Post

 **mp-units** HEAD ▾

  Search

 mpusz/mp-units
v2.5.0 ☆ 1.4k 📄 118

HomeGetting StartedTutorialsUser's GuideWorkshopsHow-to GuidesExamplesReferenceBlogRelease Notes

[← Back to index](#)

 **Mateusz Pusz**
Creator

Metadata

 November 5, 2024

 in Releases

 8 min read

Table of contents

ISQ quantities cleanup

Units equality

Portable text output

char_traits removed from fixed_string

Improved units' text output

Unicode identifiers

Convertibility with QuantityLike and QuantityPointLike entities

Symbolic constants implementation should be implementation-defined

mp-units 2.4.0 Released

A new product version can be obtained from [GitHub](#) and [Conan](#).

This release was unexpected. We planned a significant new feature to happen next, but while preparing for it and writing API Reference documentation, we made so many vital fixes and improvements that we decided they deserve a dedicated release first.

This post describes the most significant improvements while a much longer list of the changes introduced by the new version can be found in our [Release Notes](#).

ISQ quantities cleanup

Initially, we kept quantities defined in "IEC 80000-13: Information science and technology" in a standalone `iec80000` namespace, which was renamed to `iec` in the previous release. It turned out that this was incorrect. Those quantities are also a part of the ISQ. This is why, in this release, we moved all of them to the `isq` namespace (✖ **breaking change** ✖).

From now on, `iec` namespace does not provide any quantities and serves purely as a system of units definition. It contains binary prefixes (based on the powers of two) and some units introduced by IEC (e.g., `var`, `erlang`, `bit`, or `baud`).

 **Note**

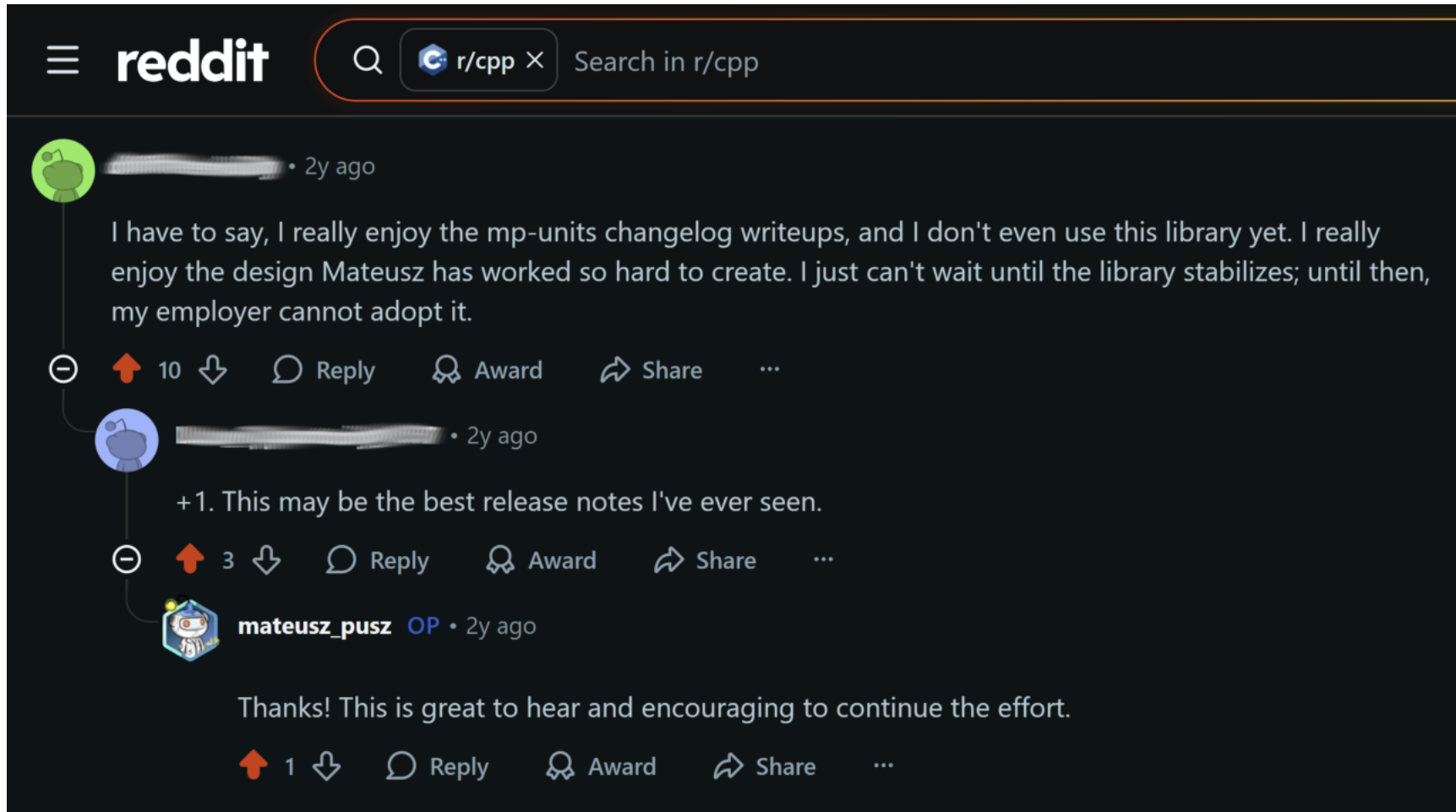
The quantities in `iec` namespace are now deprecated and will be removed in future releases.

Also, it turns out that the latest ISO 80000-3 revision makes a small cleanup to the `phase_speed` and `group_speed` quantities. Those were always defined as scalar quantities but also had alternative names `phase_velocity` and `group_velocity`. This is misleading as *velocity* is typically considered a vector quantity. It is why those `XXX_velocity` aliases were removed from the ISO

Distribution

- [LinkedIn](#) for the professional network
- [Reddit](#) ([r/cpp](#)) for the technical crowd
- [Twitter/X](#) for the "breaking news"

Distribution



TIP 9

A CULTURE OF APPRECIATION

Why Do People Contribute to Open Source?

The Reality: Open source runs on human motivation.

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WHO ACTUALLY SHOWS UP

- 1 **Your own users**, someone who hit a rough edge and fixed it
- 2 **People who care about the domain**: getting units and quantities right in C++
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Your Job: Make it rewarding enough that the few who arrive want to stay.

Building Contributor CVs (Win-Win)

The Problem: Junior developers need proof of impact.

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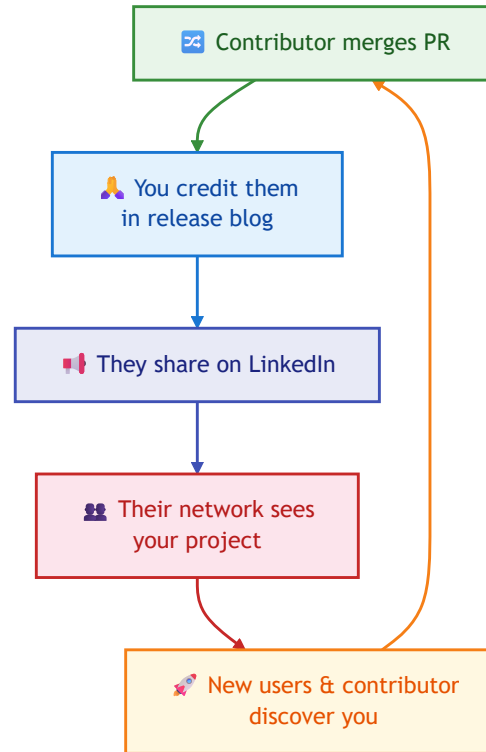
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Building Contributor CVs (Win-Win)

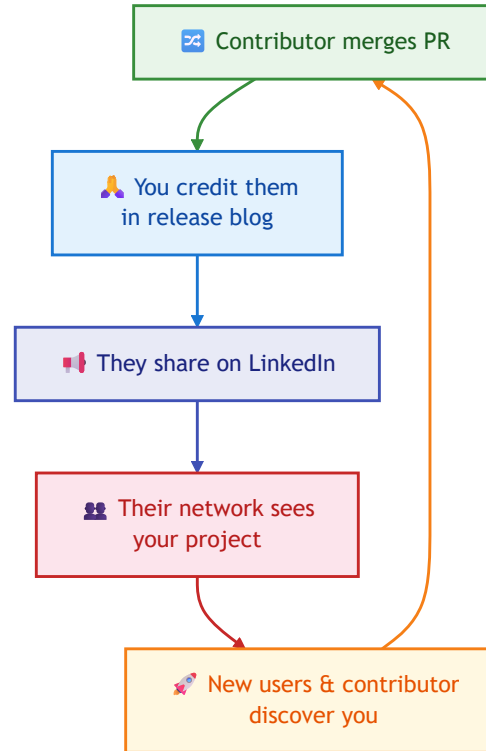
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- 3 **Testimonials:** Offer to write LinkedIn recommendations for significant contributors

The Multiplier Effect



The Multiplier Effect



People share their contributions → Free marketing for your project.

Recognition Strategies

EASY WINS

- ✓ Always say **"Thank you!"** (even for bad bug reports)
- ✓ *Mention contributors* by name in release notes
- ✓ Use GitHub's *"Co-authored-by"* for pair programming
- ✓ Create a *"Hall of Fame"* section in docs
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ADVANCED

-  *Help* contributors write blog posts about their work
-  *Invite* them to co-present at conferences
-  *Provide* reference letters for job applications

TIP 10

MANAGING THE AI INFLUX

AI Is a Tool First: Your Documentation Intern



AI Is a Tool First: Your Documentation Intern

THE WORKFLOW

- *Feed* C++ headers + design notes to an LLM
- Clearly *describe* what you need
- **Human review** for accuracy *and for voice*



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LLMS ARE GREAT WITH

- Text generation and *prose polishing*
- *Simple reasoning* and examples
- Converting bullet points to flowing text



Managing the AI Influx

THE CHALLENGE

- LLM-generated PRs are increasing



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- AI *refactors entire designs* without understanding rationale
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You need a clear AI policy



AI Contribution Policy

AI-Generated Contributions

We welcome AI assistants and coding agents (LLMs) to help you write prose, generate tests, explore a design, or draft a change. They are useful tools. But **mp-units** is a bleeding-edge C++20/23/26 library, and the bar we hold code to does not drop because a machine wrote it. Whatever tool you use, **you are the author**: you are accountable for every line, and we review it as your work.

1. **Understand and defend every line:** Never open a "blind" pull request. If you cannot explain why a change is correct and why it is designed the way it is, it is not ready, however confident the model sounded.
2. **Build and test it yourself first:** Run the change in the provided `.devcontainer` (or GitHub Codespaces) and make the tests pass locally, rather than leaving that to CI or to the reviewer. AI-written tests are especially suspect: a test must encode the *intended* behavior, not lock in whatever the code currently happens to do.
3. **Keep the diff minimal:** No unrequested reformatting, renaming, or file-wide rewrites. Large, monolithic diffs for a small change will be rejected.
4. **Watch for confident-but-wrong code:** Models are weakest exactly where this library lives, on the newest language features, template metaprogramming, and compile-time interfaces. Check for legacy patterns, needless template machinery, and subtly incorrect metaprogramming.
5. **Be transparent:** If a coding agent or an LLM did a significant part of the work, say so in the pull request, so reviewers know where to look hardest.

The "Primary Author" Principle

Protect Design Integrity.



The "Primary Author" Principle

Protect Design Integrity.

✗ **Don't:** Replace a class design just because AI suggested it

✓ **Do:** Ask the primary author about their rationale first



The "Primary Author" Principle

REMEMBER

- The original design often had **specific reasons**
 - *Compile-time performance* considerations
 - *Edge cases* the AI doesn't see
 - *Consistency* with the broader architecture



The "Primary Author" Principle

REMEMBER

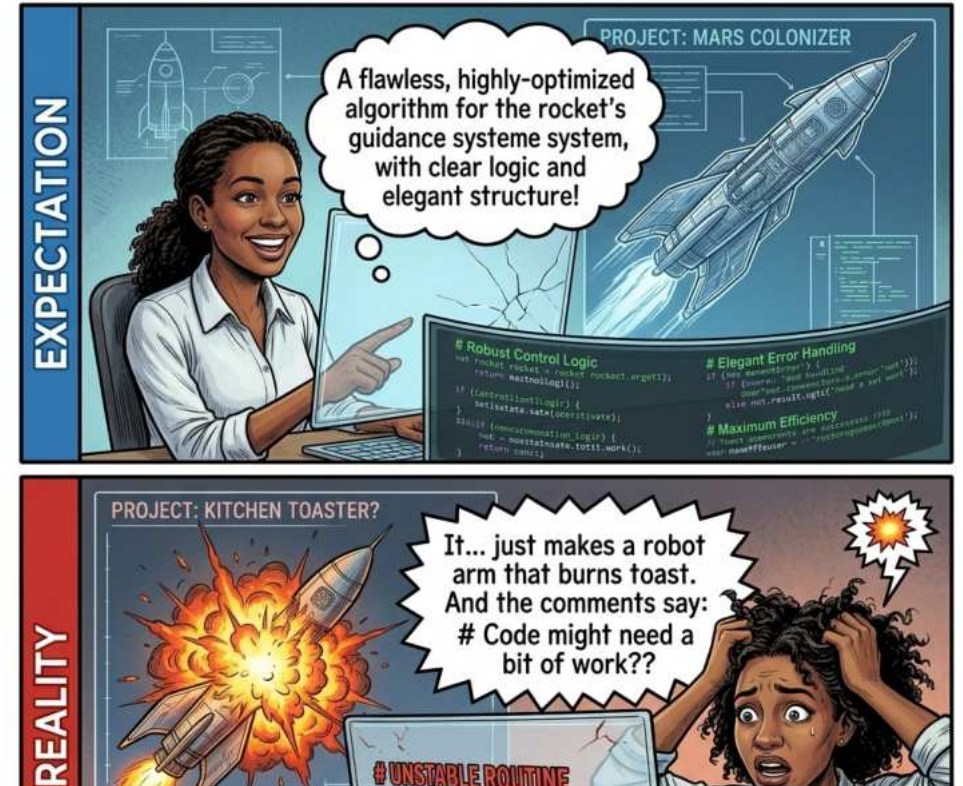
- The original design often had **specific reasons**
 - *Compile-time performance* considerations
 - *Edge cases* the AI doesn't see
 - *Consistency* with the broader architecture
- Most of them were **never written down**
 - *Approaches* you tried and dropped
 - *Discussions* that settled the design
 - What you *learned since*



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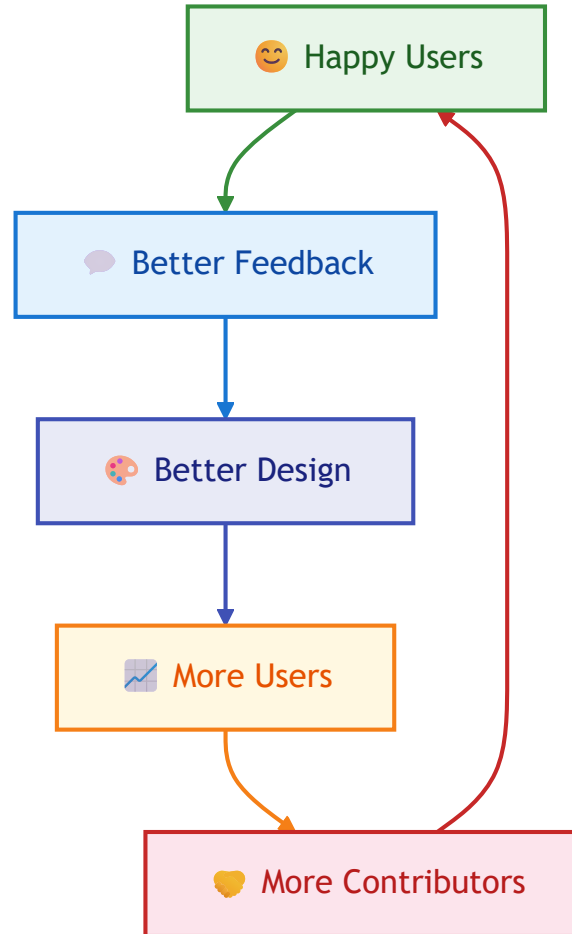
A model reads the code. It cannot read the experience that produced it.

Building a Welcoming Culture

THE FOUNDATION

- **Be respectful** to all users and contributors
- **Assume good faith**, never that the user is holding it wrong
- When someone hits a mistake, fix what misled them (a confusing name, a missing precondition): treat it as a bug in the library
- **Empathy over ego**; every issue is production feedback you would otherwise pay for
- Follow GitHub's [Community Standards](#)

The Feedback Loop



TIP 11

PROACTIVE COMMUNITY BUILDING

Proactive Community Building

Passive maintenance leads to project death. You must be the "Chief Evangelist".

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If you don't talk about your project, who will?

Scaling the Niche

Niche projects (like **mp-units**) need education.

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Write about the domain (physics, math), not just the code.

Seeking Production Feedback

- Create an issue: "I'm looking for production users."
- Be prepared for *complaints rather than praise*
- Listen to the *"boring" problems*
 - compilation time
 - header bloat
 - ...

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Always thank for feedback! Learn and improve from it.

Talk to the Community

- 1 Conference talks (like this one!)
- 2 Local meetups
- 3 Discord / Slack participation

SUMMARY

The Professional Product Checklist

STAGE 1: DISCOVERY

-  **Tip 1:** Unique naming, great README

STAGE 2: EVALUATION

-  **Tip 2:** Green master, matrix CI

STAGE 3: UNDERSTANDING

-  **Tip 3:** Diátaxis documentation

STAGE 4: INTEGRATION

-  **Tip 4:** Compiler Explorer links
-  **Tip 5:** CMake + Conan/vcpkg

STAGE 5: CONTRIBUTION

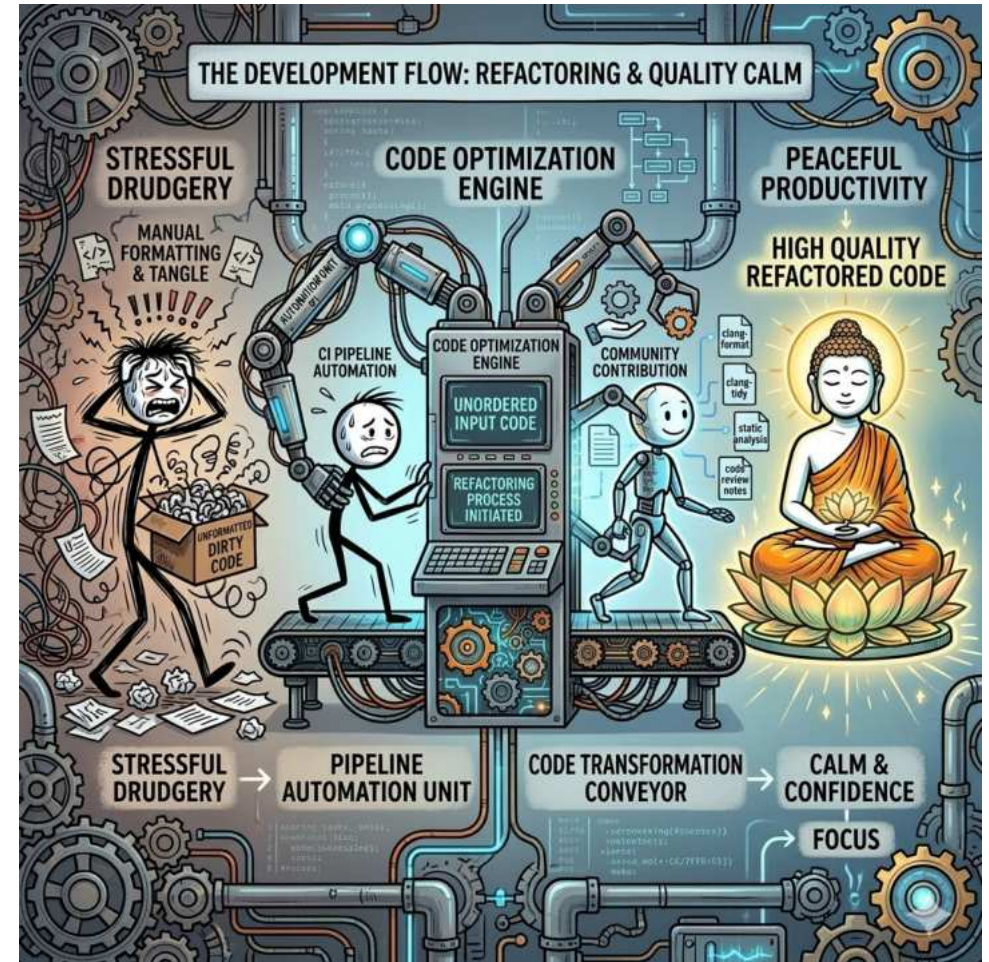
-  **Tip 6:** Issues, Discussions, empathy
-  **Tip 7:** GitHub Codespaces

STAGE 6: COMMUNITY

-  **Tip 8:** Release blogs, not just tags
-  **Tip 9:** Build contributor CVs
-  **Tip 10:** AI policy + LLM doc drafting
-  **Tip 11:** Proactive evangelism

Avoiding Maintainer Burnout

You Can't Do It All - And You Shouldn't.

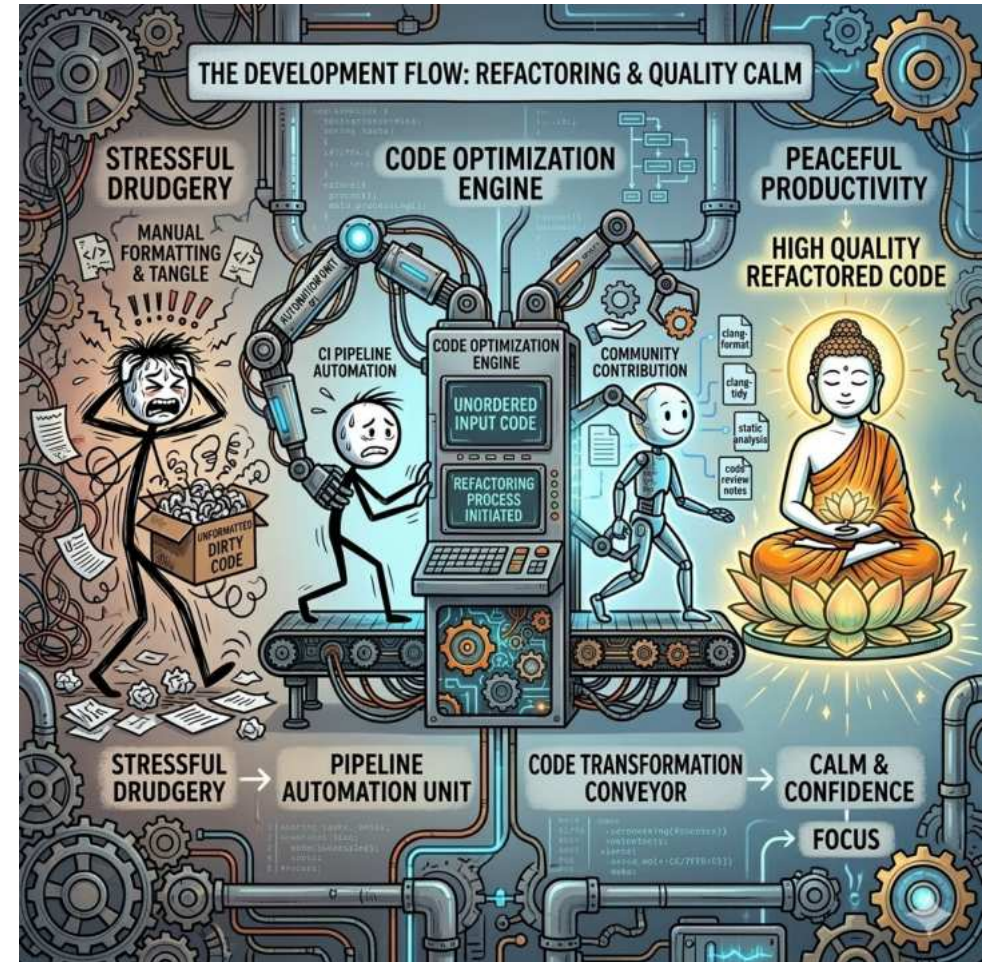


Avoiding Maintainer Burnout

You Can't Do It All - And You Shouldn't.

AUTOMATE THE BORING

- *CI* handles builds and tests
- ☒ *Clang-format* handles code style debates
- ☒ *Bots* handle stale issues
- ☒ *Release scripts or AI* generate changelogs

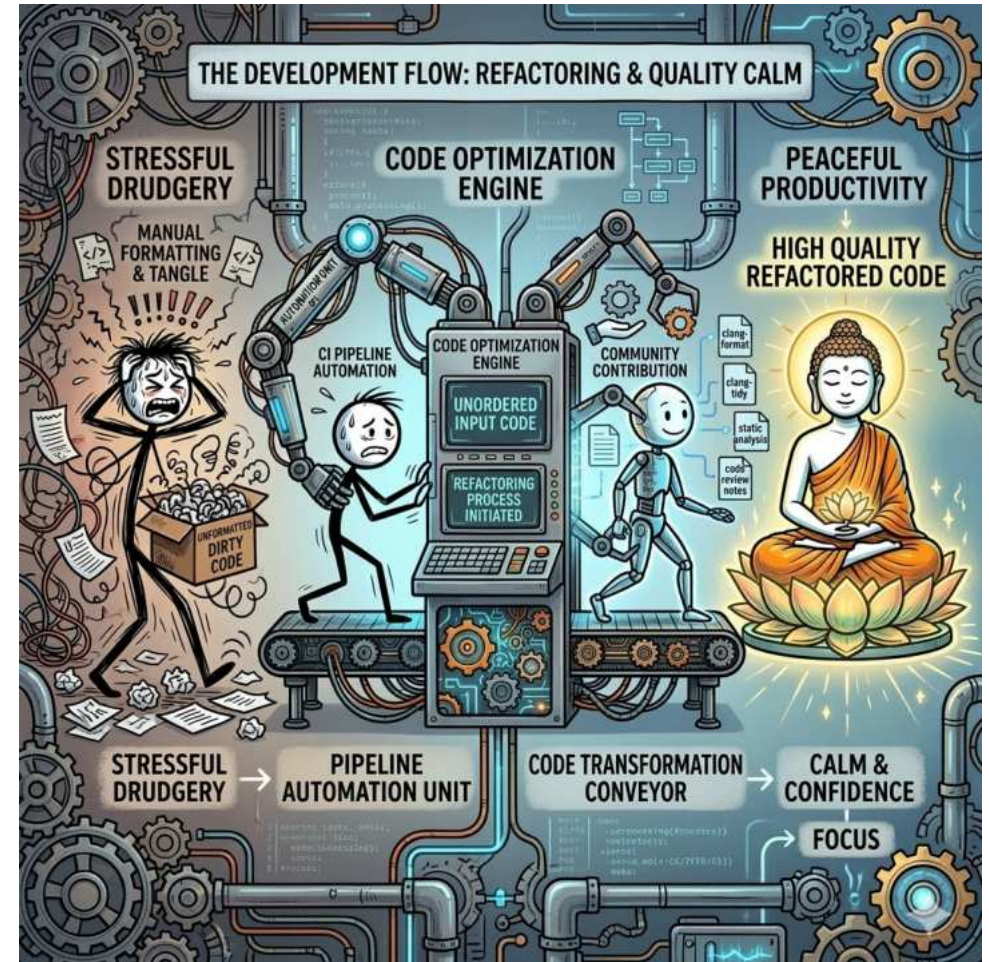


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EMPOWER OTHERS

- Give *triage permissions* to active contributors
- ☒ Document your *decision-making process*
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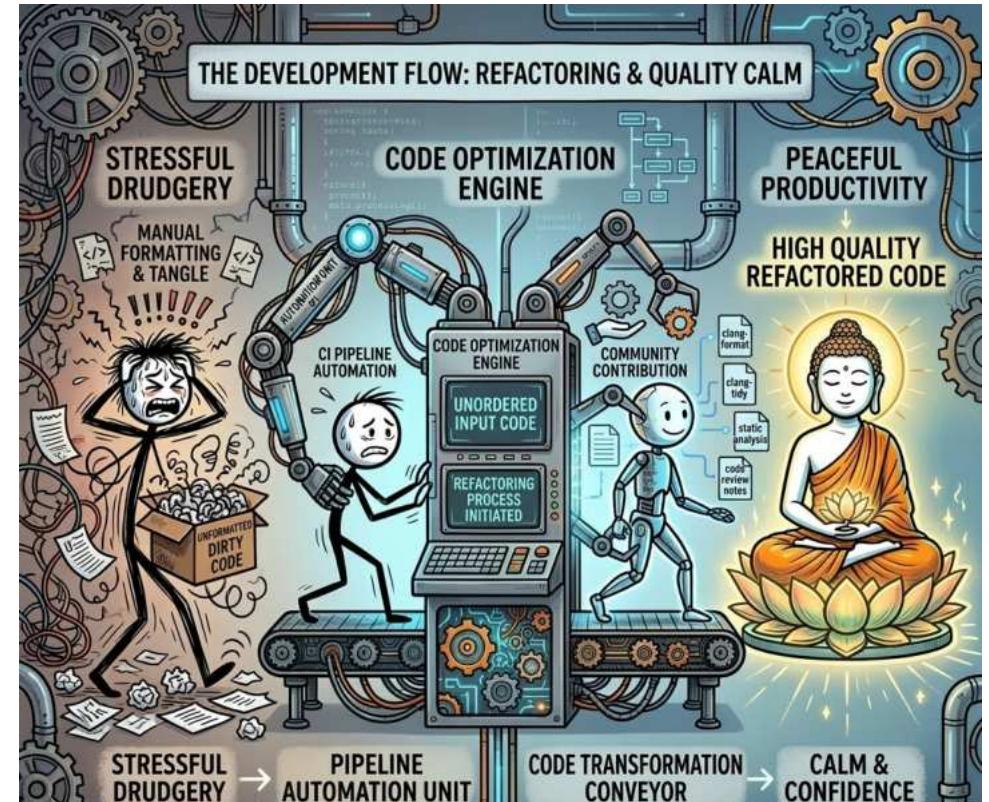
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Protect your time and energy.



Advice for Maintainers (Authors)

Be the maintainer you wish you had!

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WHEN REVIEWING PRS

- 🙏 Always thank the contributor first
- 💡 Explain your design decisions, don't just reject
- 🎯 Guide them to success, don't just point out failures
- 📄 When asking for changes, link to examples

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Remember: Every "no" without explanation discourages future contributions.

From Fortress to Forum



BEFORE: THE FORTRESS

- Hard to discover
- Brittle builds
- Hostile to questions
- Doxygen dictionary
- "Figure it out yourself"

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- Hard to discover
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AFTER: THE PROFESSIONAL PRODUCT

- Searchable and branded
- Rock-solid CI
- Welcoming community
- Prose documentation
- Clear contributor path

Don't Delay Writing Docs forever!



The Motivation: Why Bother?

YOU SHOULD, BECAUSE

- 1 **Self-Reflection:** Writing docs reveals API design flaws
- 2 **Better Design:** If it's hard to explain, the API is probably wrong
- 3 **Recognition:** Your hard work deserves users
- 4 **Wider Adoption:** Bleeding-edge code is only useful if people compile it
- 5 **Legacy:** Build something that outlives your interest

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Better Docs = Better Code.

Call to Action

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Remember: Repo → Product is a journey, not a sprint.



The background is a solid yellow color. It is decorated with several black geometric shapes, primarily parallelograms and triangles, arranged in a pattern that suggests a 3D perspective or a stylized architectural design. These shapes are positioned around the edges and corners of the frame.

CAUTION
Programming
is addictive
(and too much fun)